JEPA-Anything, anotado

JEPA-Anything: Learning Predictive Models across Different Worlds · T. Cui et al. (2026) · arXiv:2609.20800

Para qué son estas notas. Acompañan la semana 2 de la planificación (sesiones S4 a S6). Supone leído I-JEPA. Como el paper es a una columna, el **margen derecho** lleva lo esencial y lo crítico (CLAVE, PREGUNTA, CUIDADO) y el **izquierdo**, el apoyo (APUNTE, EJERCICIO, PROFUNDIZAR, EXTRA).

Si te quedas con cinco ideas

1. OPF rota el espacio objetivo, lo parte en K subespacios ortogonales con un predictor cada uno y recompone el estado con la pseudoinversa. En el límite ortogonal exacto, la pérdida de predicción es **la misma** que la del JEPA monolítico (pág. 7): lo que cambia es la arquitectura por ramas y los regularizadores.
2. La ortogonalidad garantiza que recomponer no amplifica errores ($\kappa = 1$), pero no los corrige: no es el mecanismo M2 de un latente discreto. En los rollouts, el error crece igual o más rápido que en el JEPA estándar (tabla 6).
3. En lo composicional (Pong, combinaciones de intervenciones nunca vistas) la mejora es menor que en lo ya visto: 12.9 % frente a 34.8 % (figura 4).
4. $K = 4$ casi siempre y nunca se comparan valores de K (§3.3.3, tabla 10). "Refinar o ampliar" queda fuera; los autores lo dejan como trabajo futuro (pág. 25).
5. Un factor todavía no es una regla: es continuo, se lee con $R^2 \approx 0.3$–0.4 y no tiene nombre. Las aplicaciones "científicas" (fármaco, Kepler) son las menos documentadas.

Ruta por sesión. S4: págs. 1–6 y 9. S5: págs. 6–8, 16 y 29. S6: págs. 10–25 (prioridad: 14–16, 18–19, 21–22 y 25) y 30. Las páginas 25–28 son referencias.

Colores de las notas

CLAVE	Idea esencial: lo que conviene recordar del paper.
PREGUNTA	Conexión con P1, P2, los mecanismos M1–M6 o las preguntas C1–C10 del .epub.
PROFUNDIZAR	Para más adelante: lecturas, experimentos que faltan, extensiones posibles.
CUIDADO	Límites de la evidencia, comparaciones dudosas, erratas.
APUNTE	Concepto previo o notación que el paper da por sabida.
EJERCICIO	Algo que conviene verificar a mano.
EXTRA	Detalle que se puede saltar en una primera lectura.

Cómo leer. Cada nota tiene un número. El mismo número aparece junto a una **barra de color** al lado del pasaje del original, y una línea fina las une. Si la nota se refiere a una ecuación o figura precisa, esta aparece **enmarcada** con el mismo color.

Etiquetas usadas en las notas

Remiten a *Discretizar, conjeturar, refinar* (abreviado **ep.**), y las sesiones a la planificación de lectura.

P1	¿Por qué, al discretizar, mejoran los algoritmos? ¿Qué principios son comunes?
P2	¿Cómo se modelan las reglas? ¿Qué genera las hipótesis? ¿Cuándo refinar y cuándo ampliar?
Mecanismos (ep. §3)	
M1	Compresión y navaja de Occam (§3.1)
M2	Corrección de errores y estabilidad (§3.2)
M3	Multimodalidad: elegir en vez de promediar (§3.3)
M4	Combinatoria sin mezcla, principio particulado (§3.4)
M5	Búsqueda y enumeración (§3.5)
M6	Comunicación y transmisión (§3.6)
§3.7	Lo que se pierde al discretizar
Preguntas concretas (ep. §7)	
C1	¿Discretización, cuello de botella o largo variable?
C2	¿Qué mecanismo actúa en cada caso?
C3	¿Dónde está el nivel básico?
C4	¿Cuántos factores necesita un mundo?
C5	¿Es un factor una regla?
C6	¿De dónde salen las hipótesis en cada modelo?
C7	¿Cuándo pasar de refinar a reemplazar?
C8	Varias tareas: ¿reutilizar o crear?
C9	¿Qué parte es innata?
C10	¿Qué hace el sueño con las reglas?

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JEPA-Anything: Learning Predictive Models across Different Worlds

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Code: JEPA-Anything | **Models: Jepa-Anything**

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Abstract

World modeling enables intelligence to anticipate consequences, guide interventions, and learn from interaction. Yet predictive models remain domain-specific: *can a common learning principle support world modeling across radically different systems?* We introduce **JEPA-Anything**, a domain-agnostic framework based on *orthogonal predictive factorization* (OPF). Extending joint-embedding predictive architectures, OPF decomposes latent targets into complementary factors, learns them through dedicated pathways, and recombines them within a shared predictive design. We evaluate JEPA-Anything across **seven domains**: vision, biology, clinical trajectories, control, molecular dynamics, physical fields, and weather. Experiments span representation learning, intervention prediction, out-of-distribution generalization, and long-horizon dynamics, including **10 matched dynamics tasks**, forecasting of **over 1,000 clinical events**, and **100-step molecular rollouts across four systems**. Against matched JEPA baselines, JEPA-Anything improves reported metrics on all 10 dynamics tasks and reduces single-intervention prediction error on Interventional Pong by **34.8%**. It achieves the lowest one-step and 100-step molecular errors among compared methods in all four systems. Beyond prediction, a factor-nominated biological intervention receives experimental support in cell co-cultures, patient-derived organoids, tumor fragments, and mice; latent orbital modes recover the Keplerian scaling exponent with a fitted slope of −1.4991. These results support a common factorized predictive principle across heterogeneous worlds, connecting world modeling with intervention and experimentally grounded scientific discovery.

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1 CLAVE Pregunta del paper: ¿un **mismo principio** de aprendizaje predictivo sirve para mundos tan distintos como imágenes, células, pacientes y clima?

2 CLAVE La pieza nueva es **OPF**: partir el objetivo latente en factores **ortogonales**, uno por predictor, y recomponerlos. JEPA-Anything = JEPA + OPF + un adaptador por dominio.

3 EXTRA Siete dominios y tres grupos de experimentos. No hace falta leerlos todos: la ruta está en la portada.

4 CUIDADO "Mejora en las 10 tareas": varias mejoras son mínimas (DMC píxeles, 0.4 %, figura 5) o están dentro del ruido (control en CausalWorld). Lee los números, no el resumen.

5 CUIDADO Los resultados "científicos" (fármaco, Kepler) son los más llamativos y los menos documentados: no se describe el método de nominación y Kepler es una sola corrida (§3.4).

6 PREGUNTA Partir el espacio en piezas es discretizar **la estructura**, no los valores (ep. §1 y tabla 2). Lee el paper preguntando qué gana eso (P1).

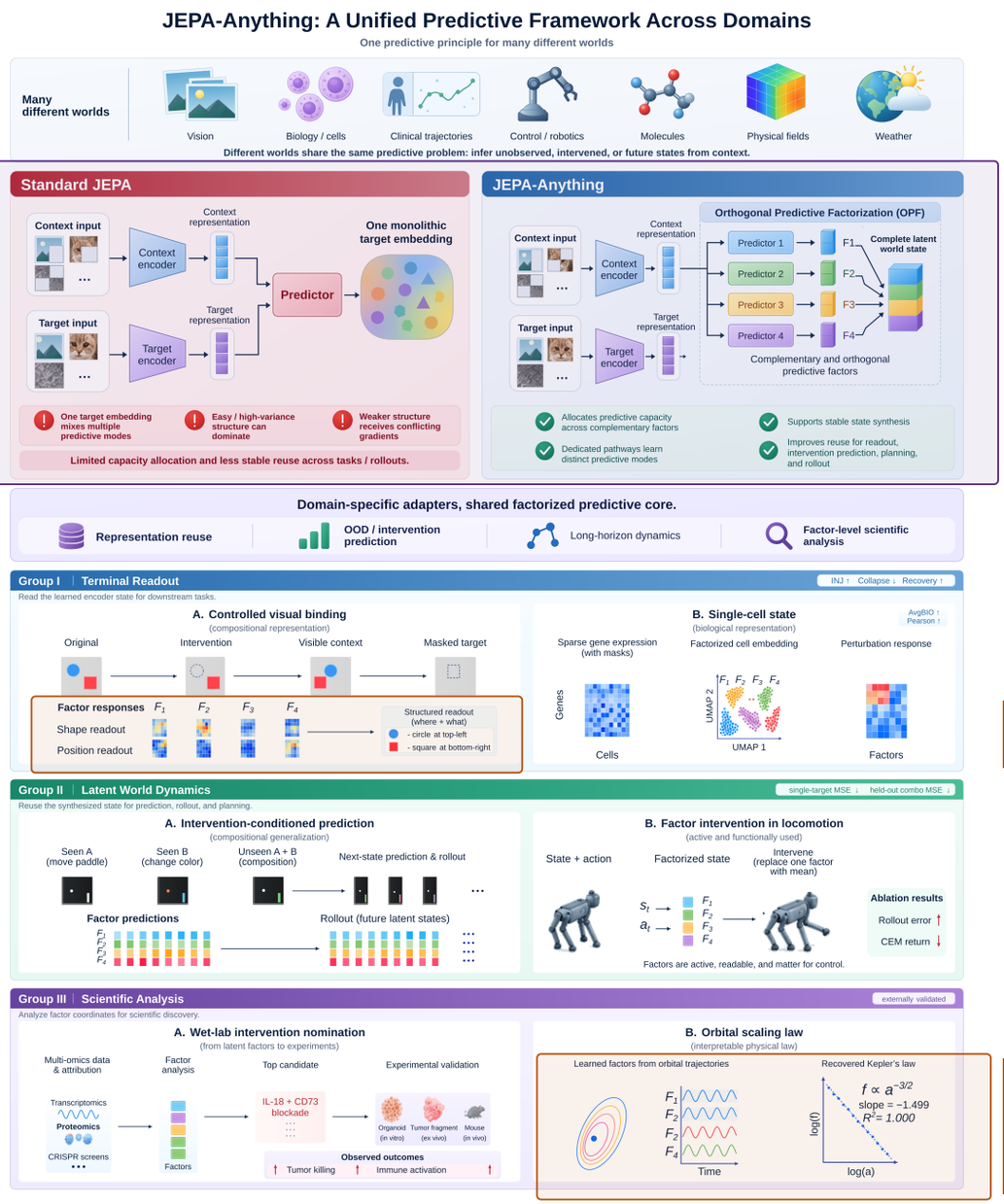


Figure 1 | **Overview of JEPA-Anything. Upper section:** Diverse domains share the same predictive problem: inferring unobserved, intervened, or future states from context. Standard JEPA mixes multiple predictive modes in a monolithic target embedding. In contrast, JEPA-Anything uses orthogonal predictive factorization (OPF) to allocate predictive capacity across complementary, orthogonal factors, each with a dedicated predictor, and synthesize a complete latent world state. The shared factorized core supports representation reuse, out-of-distribution and intervention prediction, long-horizon dynamics, and factor-level scientific analysis. **Lower section:** Three evaluation groups illustrate these capabilities. Group I (terminal readout) examines controlled visual binding and single-cell state representations. Group II (latent world dynamics) evaluates intervention-conditioned prediction and factor intervention in locomotion. Group III (scientific analysis) investigates wet-lab intervention nomination and the recovery of an orbital scaling law. Within each group, experiments A and B are arranged from left to right.

1 **APUNTE** Los recuadros rojos son los problemas que atribuyen al JEPA estándar: un embedding mezcla modos predictivos, lo fácil domina y lo débil recibe gradientes en conflicto (pág. 3).

2 **CUIDADO** El esquema sugiere factores limpios (“forma”, “posición”). Los experimentos no muestran esa nitidez: en locomoción cada factor explica $R^2 \approx 0.3$–0.4 de una variable (§3.3.4). La figura es una idealización.

3 **CUIDADO** “$R^2 = 1.000$” para Kepler: viene de una sola corrida y sin comparación con un JEPA estándar (§3.4.2).

4 **CLAVE** **La figura 1 lo resume todo.** Arriba: JEPA estándar (un predictor y un embedding “monolítico”) frente a JEPA-Anything (4 predictores, 4 factores y un estado completo recompuesto).

5 **PREGUNTA** Grupo III: la promesa de usar factores para **descubrir** (abducción, ep. §2.2). Pregunta para S6: ¿descubren algo nuevo o reconocen algo ya sabido? (C5, C6).

1. Introduction

1 APUNTE Modelo del mundo: un estado interno que permite anticipar consecuencias (Craik, ep. §2.5). Cita a World Models y a DreamerV3, que ya viste en el curso.

World models learn internal states that allow an agent or scientific model to anticipate unobserved consequences of an observed situation. Early latent world models compressed observations and learned recurrent dynamics for control [12]; more recent systems have shown that learned latent dynamics can support planning across diverse control domains [14]. Across these formulations, the central object is a predictive state that retains the information needed to anticipate how the modeled system can change.

This perspective extends beyond action-conditioned control. A predictive state may summarize a physical configuration, a patient history, a molecular trajectory, a partially observed scene, or a cellular profile. The requested target may be a future state, a hidden spatial region, another view, or a more complete observation of the same system. We therefore use *latent world model* in a broad but operational sense: a model that constructs a latent state from context and uses it to predict another state of the same underlying world. A meaningful context–target relation links the observed context to another state of the same system, and the learned state supports reuse across target queries, task-specific readouts, or repeated state transitions.

2 CLAVE Definición operativa amplia: un modelo que construye un estado latente desde un contexto y predice **otro estado del mismo mundo** (futuro, región oculta u otra vista).

Within this shared interface, each system retains its own predictive structure. A visual state may combine location, object, and transformation; a biological state may combine cell identity and response; a physical state may combine entities, scales, and dynamical modes. The number, scale, and difficulty of these predictable components vary across domains, and different downstream tasks may reuse different mixtures of them. Some targets are relatively focused, such as a local masked region or the short-term effect of a single intervention. Others, such as an unseen combination of interventions, a future patient state supporting many event risks, or a full weather field or molecular configuration, involve multiple entities, variables, and scales with unequal predictability.

3 CLAVE Hipótesis de partida: cada dominio tiene su propia **estructura predictiva** (lugar, objeto y transformación; identidad celular y respuesta…), con componentes de distinta escala y dificultad.

4 PREGUNTA "El número, escala y dificultad de estos componentes varía entre dominios"… y aun así usan $K = 4$ casi siempre (tabla 10). La motivación pide un K adaptativo que el método no tiene (C4).

5 APUNTE JEPA en cuatro líneas: context encoder, target encoder y predictor; predecir en latente abstrae el ruido y la textura. Es lo que viste en I-JEPA.

Joint-embedding predictive architectures (JEPAs) provide a natural mechanism for learning such states [3, 7]. A context encoder summarizes what is observed, a target encoder defines the state to be predicted, and a predictor maps between the two in representation space. Latent prediction allows the model to focus on shared, predictable structure while abstracting away noise, local texture, and other raw-space details that may be irrelevant to downstream use. Reconstruction can overemphasize high-variance directions that are weakly aligned with perceptual usefulness [5], whereas the learned target encoder determines the abstraction level of a JEPA target. This view also connects latent prediction to energy-based learning [20], learned similarity metrics [8, 13], masked image prediction [3], and video representation learning [7].

6 PROFUNDIZAR Ref. 5, Balestriero y LeCun (2024): por qué reconstruir píxeles da rasgos poco útiles (prioriza direcciones de alta varianza). Buena lectura para M1 y P1.

JEPA supplies the common latent-prediction mechanism, but the usual formulation still represents the requested world state through one target embedding and one prediction pathway. When local and global structure, multiple entities, or changes at different scales share this monolithic target, easily predicted or high-variance structure may dominate optimization, several latent directions may serve similar roles, and weaker predictive structure may receive conflicting gradients. This makes latent world-state design a predictive capacity-allocation problem: the shared interface should remain fixed while the state can be organized into multiple complementary components.

7 CLAVE **Diagnóstico central (S4).** Con un objetivo monolítico, lo fácil o de alta varianza domina el gradiente, varias direcciones cumplen el mismo papel y la estructura débil recibe gradientes en conflicto. Lo plantean como un **problema de asignación de capacidad**.

8 CUIDADO Este diagnóstico se **afirma**, pero no se mide en un JEPA estándar (con el espectro de la representación o con gradientes en conflicto). Se infiere de que OPF mejora.

Can one latent world-model interface organize the differently structured predictive states of many context–target systems through multiple factors?

We answer this question with JEPA-Anything, a latent world-modeling framework based on *orthogonal predictive factorization*. A collection of learned basis matrices analyzes each target state into multiple components, and a dedicated branch predicts each component from the shared context representation. Within-factor and cross-factor orthogonality objectives allocate different directions of the target space to different branches. Factor-activity regularization maintains variation in projected targets, while an online variance term discourages collapse of the trainable encoder. The predicted components can be synthesized into a complete latent state for decoding, planning, or autoregressive rollout. Factor identities emerge from predictability, and the factors jointly shape the complete online state. Ordinary downstream tasks read from that encoder state, whereas operational transitions and factor-level analyses retain and combine the explicit factor coordinates. The number and width of factor blocks can be configured to the predictive complexity of an instantiation under the same learning principle and latent world-state interface.

The word *Anything* denotes the breadth of the latent world-state interface. Each domain defines its observation tokens, context–target views, structural descriptors, and encoder; once the context and target states are available, the same factorized predictive core applies. Temporal conditioning, masking, and partial observation instantiate future, spatial, biological, and other structured targets. Adapters define observation geometry, OPF organizes predictive capacity, and readout, synthesis, or factor analysis determines how the learned state is used.

We evaluate this argument at three levels. First, visual binding and single-cell tasks read the online encoder state, while longitudinal health reads a one-step synthesized future state; in all three, evaluation ends at a task-specific readout. Second, intervention-conditioned sequences, control, molecular dynamics, physical fields, and weather test whether the synthesized state supports compositional prediction and repeated transition. Third, biological and orbital analyses test whether the factor coordinates provide a reusable diagnostic interface. The results improve readout quality across vision, cells, and disease forecasting; reduce intervention-prediction error on both observed and unseen combinations; improve the reported metrics across the ten-task matched dynamics benchmark and all four molecular systems; and connect learned coordinates to wet-lab evidence and a known physical scaling law.

Figure 1 summarizes the shared predictive architecture and the three evaluation modes used throughout the paper.

Our contributions are summarized as follows:

- We define latent world modeling through a common context–target state interface: domain adapters handle observation geometry, while a modality-independent core organizes the resulting predictive states.

1 **CLAVE** Receta completa: bases aprendidas P_k que proyectan el objetivo, una rama que predice cada componente, ortogonalidad dentro de cada factor y entre factores, un piso de actividad por factor, varianza del encoder contra el colapso y síntesis del estado completo.

2 **PREGUNTA** "Las identidades de los factores emergen de la predictibilidad": no se nombran de antemano. Eso los acerca a conceptos aprendidos, pero no garantiza que sean **reglas** (discretas, nombrables, componibles; C5).

3 **PREGUNTA** "El número y el ancho de los bloques se configuran": K es un hiperparámetro **fijo**. Nunca se decide crear o fusionar factores según los datos; "refinar o ampliar" queda fuera (ep. §4.4; C4, C7).

4 **PREGUNTA** Lo "de fábrica" en cada dominio: tokenización, descriptores, qué es contexto y qué es objetivo, y la arquitectura del encoder. El núcleo OPF es lo único común (ep. §4.3, C9).

5 **EXTRA** Mapa de los experimentos: el grupo I lee el encoder, el grupo II reutiliza el estado predicho en bucle y el grupo III analiza los factores. Para nuestras preguntas, pesa más el grupo II.

- We introduce orthogonal predictive factorization, which partitions a latent target into learned subspaces with dedicated predictors, providing configurable predictive capacity and a complete state for reuse.
- We combine factorized prediction with within- and cross-subspace orthogonality, factor activity, and online encoder variance, yielding a common interface for stable state synthesis and factor-level diagnostics.
- We instantiate the same core across physical-field, weather, molecular, biological, visual, clinical, and control systems, evaluating readout, OOD forecasting, planning, rollout stability, and scientific diagnostics.

2. JEPA-Anything

2.1. One Interface, One Predictive Core

The unifying object in JEPA-Anything is a latent world-state interface: structured observations specify an available context, a requested target, and any target descriptors. The top row of Figure 1 shows this shared architecture. Adapters expose a domain's predictive state, while OPF distributes its capacity across complementary coordinates, allowing observation geometry and predictive complexity to vary without changing the interface.

1 **APUNTE** Notación. H: tokens de contenido; S: descriptores estructurales (posición, tiempo, identidad); C y T: índices de contexto y objetivo.

1 Let δ index a domain and let $x \sim \mathcal{D}_\delta$ be a raw observation. A domain adapter $\mathcal{A}_\delta$ maps x into content tokens $H = \{h_i\}_{i\in\Omega}$ and structural descriptors $S = \{s_i\}_{i\in\Omega}$ over an index set Ω. A descriptor may encode a patch coordinate, time stamp, graph position, entity identity, or may be empty when no additional structure is needed. A view sampler $\mathcal{V}_\delta$ selects context indices $C \subset \Omega$ and target indices $T \subset \Omega$. This gives the common interface

$$x \xrightarrow{\mathcal{A}_\delta} (H, S) \xrightarrow{\mathcal{V}_\delta} (H_C, S_C, T, S_T). \tag{1}$$

2 **CLAVE** **Ecuación (1).** Todo lo específico del dominio vive aquí: el adaptador $\mathcal{A}_\delta$ produce tokens y descriptores, y el muestreador $\mathcal{V}_\delta$ elige contexto y objetivo. Lo demás es un solo algoritmo.

The semantics of $\mathcal{A}_\delta$ and $\mathcal{V}_\delta$ are domain-specific; everything after Equation (1) follows one algorithm.

3 **APUNTE** $z_c = f_\theta(H_C, S_C)$ es el contexto. z_t se calcula con **toda** la observación y se recorta después, igual que en I-JEPA (su tabla 11).

3 An online encoder produces a context representation $z_c = f_\theta(H_C, S_C)$. A target encoder produces a latent target $z_t = f_{\bar\theta}(H, S)_t \in \mathbb{R}^d$ for every $t \in T$. Using a full target view includes the familiar JEPA case in which target tokens are selected after encoding. The target parameters are updated as an exponential moving average,

4 **EXTRA** EMA, igual que en I-JEPA.

$$\bar\theta \leftarrow m\bar\theta + (1-m)\theta, \qquad 0 \le m < 1, \tag{2}$$

and receive no gradients. The single-vector formulation is recovered when $|T| = 1$.

Table 1 states the boundary precisely. Hyperparameters such as latent width or number of factors may be selected for computational scale; the functional form of the additive OPF objective is unchanged.

5 **CUIDADO** "Hiperparámetros como el número de factores pueden elegirse por escala computacional": admiten que K es una perilla de ingeniería, no algo inferido de los datos.

Table 1 | What changes and what remains invariant across JEPA-Anything instantiations.

Component	Domain-specific boundary	Shared JEPA-Anything core
Raw representation	Image, sequence, graph, set, record	—
Tokenization and descriptors	$\mathcal{A}_\delta$	Common token interface
Context–target semantics	$\mathcal{V}_\delta$	Context predicts targets
Encoder architecture	ViT, Transformer, GNN, MLP, . . .	Online/EMA-target scheme
Predictive learning	—	Factorized predictors and state synthesis
Regularization	—	Orthogonality, factor activity, encoder variance
Training objective	Original loss $\mathcal{L}^{(\delta)}_{\text{base}}$	Additive OPF loss in Equation (10)
Downstream use	Pooling, readout, probe, decoder, planner	Online-encoder features or synthesized predicted states

1 **PREGUNTA** **Tabla 1 = la frontera entre lo innato y lo común.** Columna central: lo que el diseñador aporta por dominio. Columna derecha: el principio general (ep. §4.3, C9).

The interface requires a meaningful predictive relation: the target must have usable statistical dependence on the context and represent another state of the same underlying system. It covers pixels, trajectories, graphs, sets, fields, and multivariate records; factor identities emerge from predictive learning rather than predefined semantics.

2 **CLAVE** Condición de uso: el objetivo debe depender estadísticamente del contexto y ser otro estado del **mismo** sistema. Sin relación predictiva no hay nada que aprender.

2.2. Orthogonal Predictive Factorization

Standard JEPA training predicts a target embedding through a single predictor. We instead introduce K learned projectors $P_k \in \mathbb{R}^{d\times r}$ and choose r such that $Kr = d$. In the zero-penalty limit, the resulting mutually orthogonal subspaces form a full partition of the target space. We factorize the stop-gradient target representation as

$$\widetilde{z}_t = \mathrm{sg}(z_t), \qquad z_t^{(k)} = P_k^\top \widetilde{z}_t, \qquad k = 1, \ldots, K. \tag{3}$$

3 **CLAVE** K proyectores $P_k \in \mathbb{R}^{d\times r}$ con $Kr = d$: los K subespacios ortogonales **particionan** todo el espacio objetivo en piezas que no se solapan.

4 **APUNTE** $z_t^{(k)} = P_k^\top \tilde{z}_t$: las coordenadas del objetivo en la base del factor k. $\tilde{z}_t$ lleva **stop-gradient**: el objetivo no se mueve con esta pérdida.

Stop-gradient is applied to the target-encoder output, while the projectors remain trainable and receive gradients from the predictive, orthogonality, and factor-activity terms. The learned subspaces are therefore predictive and orthogonality-regularized.

5 **APUNTE** Los P_k sí reciben gradiente: el modelo elige **qué direcciones** van a cada factor según lo que logra predecir. Ese es el sentido de "emergen de la predictibilidad".

Each factor has a corresponding predictor q_k that maps the shared context representation and, when needed, target descriptor s_t into the same r-dimensional space,

$$\widehat{z}_t^{(k)} = q_k(z_c, s_t). \tag{4}$$

6 **APUNTE** Un predictor q_k por factor, que puede ser una red independiente o una cabeza sobre un tronco común.

The factor predictions are concatenated as $\widehat{u}_t = [(\widehat{z}_t^{(1)})^\top, \ldots, (\widehat{z}_t^{(K)})^\top]^\top \in \mathbb{R}^d$. The complete target state is synthesized through the Moore–Penrose pseudoinverse of the analysis map $P^\top$,

$$\widehat{z}_t = \left(P^\top\right)^\dagger \widehat{u}_t \in \mathbb{R}^d, \qquad P = [P_1, \ldots, P_K]. \tag{5}$$

7 **CLAVE** **Síntesis (5):** $\hat{z}_t = (P^\top)^\dagger \hat{u}_t$. Si P es ortogonal, $(P^\top)^\dagger = P$ y el estado completo es $\sum_k P_k \hat{z}_t^{(k)}$: sumar cada pieza en sus direcciones.

8 **EJERCICIO** Comprueba que si $P^\top P = I_d$, entonces $(P^\top)^\dagger = P$. Pista: la pseudoinversa de una matriz invertible es su inversa.

When P is exactly orthogonal, $(P^\top)^\dagger = P$, and Equation (5) reduces to $\widehat{z}_t = \sum_k P_k \widehat{z}_t^{(k)}$. This state is the common interface for applications that decode a future latent state or feed predictions back into a multi-step rollout. The descriptor specifies the requested target location or identity. Predictors may ignore it when the context uniquely determines the target. They may be implemented independently or as a shared trunk followed by branch-specific heads.

9 **CUIDADO** "Un tronco común seguido de cabezas por rama": si las cabezas fueran **lineales**, $P[q_1; \ldots; q_K]$ sería otra capa lineal y la factorización no cambiaría la clase de funciones, solo la parametrización. El paper no dice qué variante usa en cada caso.

For temporal, action-conditioned, or intervention-conditioned instances, let ξ_t denote the exogenous input at step t—for example, an action, intervention label, or known forcing. The domain adapter places ξ_t in the context tokens or target descriptor. When the current context is summarized by a latent state z_t, the operational transition takes the form

$$\widehat{u}_{t+1} = \left[q_1(z_t, \xi_t, s_{t+1}); \ldots; q_K(z_t, \xi_t, s_{t+1})\right], \qquad \widehat{z}_{t+1} = (P^\top)^\dagger \widehat{u}_{t+1}.$$

10 **CLAVE** **Modo dinámico:** el siguiente estado se predice por factores, se sintetiza y se vuelve a alimentar. Aquí importa la estabilidad (M2): el error de un paso entra al siguiente.

1 **EXTRA** Planificación: un planificador (CEM) puntúa trayectorias imaginadas con una recompensa. OPF solo aporta el modelo.

2 **EJERCICIO** **Ejercicio clave (S5).** Con P ortogonal: $\sum_k \left\|\hat{z}^{(k)} - P_k^\top z\right\|^2 = \left\|\hat{u} - P^\top z\right\|^2 = \|P\hat{u} - z\|^2 = \|\hat{z} - z\|^2$. Es decir, salvo la constante $1/(K|T|r)$, $\mathcal{L}_{\text{pred}}$ es **exactamente** la pérdida L2 del JEPA monolítico sobre el estado sintetizado.

1 Repeated application defines a latent rollout. A planner may score the resulting trajectory with a known domain reward or a task-specific reward readout; this division lets OPF supply the predictive state while the application supplies the reward specification.

2 To preserve both the direction and magnitude required by state synthesis, we directly regress each predicted factor to its target,

$$\mathcal{L}_{\text{pred}} = \frac{1}{K|T|r} \sum_{t \in T} \sum_{k=1}^{K} \left\| \widehat{z}_t^{(k)} - z_t^{(k)} \right\|_2^2, \tag{6}$$

where sg denotes stop-gradient on the EMA target-encoder output.

3 **PREGUNTA** Consecuencia del ejercicio: en el límite ortogonal la pérdida de predicción no cambia. La ganancia de OPF tiene que venir de (i) la arquitectura por ramas, (ii) los regularizadores de actividad y varianza o (iii) la dinámica de optimización. El paper no separa estas tres (C2).

Distinct predictive factors require an explicit diversity constraint because several projectors can otherwise learn the same target directions. We require the columns within each projector to be approximately orthonormal and different projectors to occupy approximately orthogonal subspaces,

$$\mathcal{L}_{\text{orth}} = \sum_{k=1}^{K} \left\| P_k^\top P_k - I_r \right\|_F^2 + \sum_{1 \le i < j \le K} \left\| P_i^\top P_j \right\|_F^2 . \tag{7}$$

4 **CLAVE** $\mathcal{L}_{\text{orth}}$ **(7).** Primer término: cada base es ortonormal ($P_k^\top P_k \approx I_r$). Segundo término: bases distintas son ortogonales entre sí ($P_i^\top P_j \approx 0$), lo que impide que dos factores aprendan la misma dirección.

The first term avoids degenerate bases within a factor, while the second discourages different factors from repeatedly encoding the same directions. This design is related in spirit to redundancy-reduction objectives in self-supervised learning [6, 36], but applies orthogonality directly to learned predictive target subspaces, complementing objectives defined on the statistics of a single embedding.

5 **PREGUNTA** Pariente de Barlow Twins y VICReg: reducción de redundancia (Barlow, 1961; ep. §3.1). La diferencia es que aquí la no redundancia se impone a **subespacios del objetivo**, no a las dimensiones de un embedding.

6 **EXTRA** La descomposición QR exacta solo se usa para el análisis geométrico. Los modelos entrenados usan proyectores casi ortogonales y la pseudoinversa.

6 For a strict orthogonal decomposition, the concatenated basis can be factorized as $P_{\text{raw}} = QR$, where $Q^\top Q = I_d$, and partitioned as $Q = [Q_1, \ldots, Q_K]$. The resulting coordinates $Q_k^\top z$ admit transpose synthesis $z = QQ^\top z = \sum_k Q_k Q_k^\top z$. The controlled geometry analysis uses this strict decomposition. Predictive models use the learned orthogonality-regularized projectors and pseudoinverse synthesis in Equation (5).

Let $P = [P_1, \ldots, P_K] \in \mathbb{R}^{d \times d}$. The geometric role of the constraint can be stated directly.

7 **CLAVE** **Proposición 1.** Si las bases son ortonormales y mutuamente ortogonales, P es una matriz ortogonal $d \times d$: los factores son un cambio de base y no se pierde nada ($\|P^\top z\| = \|z\|$).

8 **EJERCICIO** Demuéstrala: $P_i^\top P_j = \delta_{ij} I_r$ implica $P^\top P = I_d$; como P es cuadrada ($Kr = d$), también $PP^\top = I_d$. Luego expande $\|P^\top z\|^2$ por bloques.

8 **Proposition 1** (Orthogonal predictive decomposition)**.** *If $P_i^\top P_j = 0$ for $i \neq j$ and $P_k^\top P_k = I_r$, then the factor spaces form an orthogonal direct sum and, for every $z \in \mathbb{R}^d$,*

$$\|P^\top z\|_2^2 = \sum_{k=1}^{K} \|P_k^\top z\|_2^2 = \|z\|_2^2, \qquad z = \sum_{k=1}^{K} P_k P_k^\top z. \tag{8}$$

Thus the factors preserve all information in z up to an orthogonal change of basis. In this zero-penalty limit, the pseudoinverse synthesis in Equation (5) reduces to the corresponding orthogonal synthesis rule.

9 **PREGUNTA** Lectura crítica: aquí "factorizar" es **rotar** el espacio y agrupar coordenadas en K bloques. No se discretizan valores ni identidades; se parte la geometría (ep. tabla 2, fila "estructura").

Proof. The assumptions and $Kr = d$ give $P^\top P = PP^\top = I_d$, so P is orthogonal. The norm identity follows by blockwise expansion of $P^\top z$, and $z = PP^\top z = \sum_k P_k P_k^\top z$ gives exact reconstruction. □

Corollary 1 (Stable synthesis versus unconstrained heads)**.** *Let $u = P^\top z$ and let $e \in \mathbb{R}^d$ denote an error in the concatenated predicted factors. Under the assumptions of Proposition 1,*

$$\widehat{z} = P(u+e), \qquad \|\widehat{z} - z\|_2 = \|e\|_2, \qquad \kappa_2(P) = 1.$$

Without the cross-factor orthogonality constraint, there exist admissible projectors with repeated directions for which P is rank deficient, and full-rank projectors with nearly repeated directions for which $\kappa_2(P)$ is arbitrarily large. Consequently, unconstrained heads provide no uniform guarantee of either complete recovery or stable synthesis.

Proof. Proposition 1 makes P orthogonal, so $(P^\top)^\dagger = P$, $P(u+e) = z + Pe$, and orthogonality preserves $\|e\|_2$. Without the cross-factor constraint, duplicate directions can make P rank deficient, while nearly duplicate directions drive its smallest singular value toward zero and its condition number without bound. □

Together, Proposition 1 and the corollary establish non-overlapping coverage and stable synthesis; Table 5 reports these properties in a trained model. Subsequent experiments evaluate factor activity and domain-level interpretation.

2.3. Maintaining Factor and Encoder Activity

Orthogonality specifies how the subspaces relate to one another, while a per-factor activity floor keeps every projected target coordinate active across samples. For a mini-batch $\mathcal{B}$ and all its sampled target indices, let $\sigma^{\text{fac}}_{k,j}$ be the empirical standard deviation of coordinate j in $z^{(k)}_t$. Because the target representation is stopped, this term shapes the projectors. We also define a separate activity statistic on the online context representations. Let σ^{enc}_j denote the empirical standard deviation of coordinate j in the online context representations z_c; for token-valued contexts, the statistic is computed over all valid context tokens in the mini-batch. Standard deviations are evaluated as $\sqrt{\text{Var} + \epsilon}$. We define

$$\mathcal{L}_{\text{fac}} = \frac{1}{Kr} \sum_{k=1}^{K} \sum_{j=1}^{r} \max(0, \gamma_{\text{fac}} - \sigma^{\text{fac}}_{k,j}),$$
$$\mathcal{L}_{\text{enc}} = \frac{1}{d} \sum_{j=1}^{d} \max(0, \gamma_{\text{enc}} - \sigma^{\text{enc}}_{j}). \tag{9}$$

The first term keeps projected targets active; the second sends a direct anti-collapse gradient to the online encoder. We collect the factorized prediction and regularization terms into an additive OPF loss. If $\mathcal{L}^{(\delta)}_{\text{base}}$ denotes the loss already used by the original implementation in domain δ, the optimized training loss is

$$\mathcal{L}_{\text{OPF}} = \mathcal{L}_{\text{pred}} + \lambda_{\text{orth}}\mathcal{L}_{\text{orth}} + \lambda_{\text{fac}}\mathcal{L}_{\text{fac}} + \lambda_{\text{enc}}\mathcal{L}_{\text{enc}},$$
$$\mathcal{L}^{(\delta)}_{\text{train}} = \mathcal{L}^{(\delta)}_{\text{base}} + \mathcal{L}_{\text{OPF}}. \tag{10}$$

Each application combines its base objective with $\mathcal{L}_{\text{OPF}}$; the adapter, view sampler, encoder family, and input tokenization specialize the predictive term to the data.

3 **EJERCICIO** Toma $d = 2$, $K = 2$, $r = 1$, $p_1 = (1, 0)$ y $p_2 = (\cos\varepsilon, \sin\varepsilon)$. Muestra que $\sigma = \sqrt{1 \pm \cos\varepsilon}$ y que $\kappa_2 = \cot(\varepsilon/2) \to \infty$ cuando $\varepsilon \to 0$: el caso de "direcciones casi repetidas".

1 **CLAVE** **Corolario 1.** Con P ortogonal, un error e en los factores produce **el mismo** error en el estado: $\|\hat{z} - z\| = \|e\|$ y $\kappa_2(P) = 1$. Sin ortogonalidad, P puede ser casi singular y amplificar los errores sin cota.

2 **PREGUNTA** Responde la pregunta del .epub (§3.2): esto **no** es corrección de errores. Un estado discreto se "redondea" y vuelve a la categoría correcta; aquí el error ni crece ni se corrige (es una isometría). Evita amplificar, pero no restaura.

4 **CUIDADO** La garantía es solo para el paso **lineal** de síntesis. En un rollout, el error se compone a través de predictores no lineales, y eso la proposición no lo controla.

5 **CLAVE** **Dos pisos de varianza** (tipo bisagra). $\mathcal{L}_{\text{fac}}$ obliga a que cada coordenada de cada factor varíe entre muestras (que ningún factor "se apague"). $\mathcal{L}_{\text{enc}}$ hace lo mismo con el encoder online, contra el colapso.

6 **APUNTE** $\mathcal{L}_{\text{enc}}$ es el término de varianza de VICReg. I-JEPA no lo tenía: evitaba el colapso solo con el EMA y el predictor.

7 **CUIDADO** **Ojo con la ecuación (10).** Según §3.1, el JEPA estándar de comparación **no** recibe estos regularizadores. Cada comparación mezcla factorización con anticolapso explícito. Falta el control obvio: JEPA estándar más $\mathcal{L}_{\text{enc}}$.

8 **PREGUNTA** ¿Qué explica la mejora? Candidatos: capacidad por ramas (M1), no redundancia (Barlow), varianza anticolapso o estabilidad de la síntesis (M2). Este paper solo audita la geometría (tabla 5); no separa los mecanismos (C2).

2.4. Shared Predictive Pretraining and Task-Specific Readout

Every instantiation follows the same optimization loop: (1) adapt a raw observation into tokens and descriptors; (2) sample context and targets; (3) encode context with the online encoder and targets with the stop-gradient EMA encoder; (4) predict every orthogonal target factor; (5) update the online encoder, projectors, and predictors using the base-plus-OPF training loss in Equation (10); and (6) update the target encoder by EMA. Thus, JEPA-Anything specifies one common algorithm across domain pipelines.

Representation mode. Ordinary downstream evaluation retains the online encoder as a common source of reusable features and attaches a readout that follows the structure of the domain and task. The EMA target encoder, projectors, and prediction heads are discarded. For domain δ and task τ, we write

$$h_{\delta,\tau}(x) = R_{\delta,\tau}\left(f_\theta^\delta(\mathcal{A}_\delta(x))\right), \tag{11}$$

where $R_{\delta,\tau}$ may select a token, pool a set or sequence, preserve a time-indexed state, or implement a learned probe. Orthogonal predictive factorization therefore acts as a shared training principle that shapes a reusable encoder while allowing each modality to use its own aggregation rule.

Operational world-model mode. For future-state prediction, intervention forecasting, planning, and autoregressive simulation, the learned projectors and predictors are retained. Equation (5) then supplies the complete next latent state to a decoder, planner, or subsequent transition step. Thus the same predictive pretraining principle supports two uses: reusable encoder features for state readout and an explicit latent transition interface for predictions composed over time.

For factor-level analysis, we retain the learned projectors as a diagnostic interface. Let $z_\delta(x) \in \mathbb{R}^d$ denote the online-encoder state obtained at the same token or pooled level on which target factorization was trained. We define

$$u_k(x) = P_k^\top z_\delta(x), \qquad k = 1, \ldots, K.$$

These factor coordinates expose distinct predictive modes for domain-specific analysis, while ordinary downstream tasks use the encoder state directly. In the scientific applications below, they provide the interface for intervention analysis and spectral characterization.

3. Experiments

We evaluate JEPA-Anything across heterogeneous visual, biological, clinical, control, molecular, physical-field, and weather systems as three linked tests of the proposed design. Each system defines its own observations and context–target relation while retaining the same JEPA-Anything core. Group I evaluates terminal readout: vision and single-cell tasks read the online encoder state, while disease forecasting reads a one-step synthesized future state. Group II tests whether predicted states can be repeatedly reused in intervention-conditioned and autonomous dynamics. Group III tests whether factor coordinates can be reused for scientific analysis and external validation.

1 EXTRA El bucle de entrenamiento: seis pasos, idénticos en todos los dominios.

2 CLAVE **Modo representación:** se descartan los proyectores y los predictores; queda solo el encoder online. OPF actúa como **principio de entrenamiento** que moldea ese encoder.

3 CLAVE **Modo operacional:** se conservan predictores y proyectores para hacer rollouts. Es el modo "modelo del mundo" en el sentido de Dreamer.

4 PREGUNTA **Modo análisis:** $u_k(x) = P_k^\top z_\delta(x)$ son las coordenadas del factor k. Es lo más parecido a una "variable" que ofrece el sistema, pero es continua y de dimensión r, no un símbolo (C5).

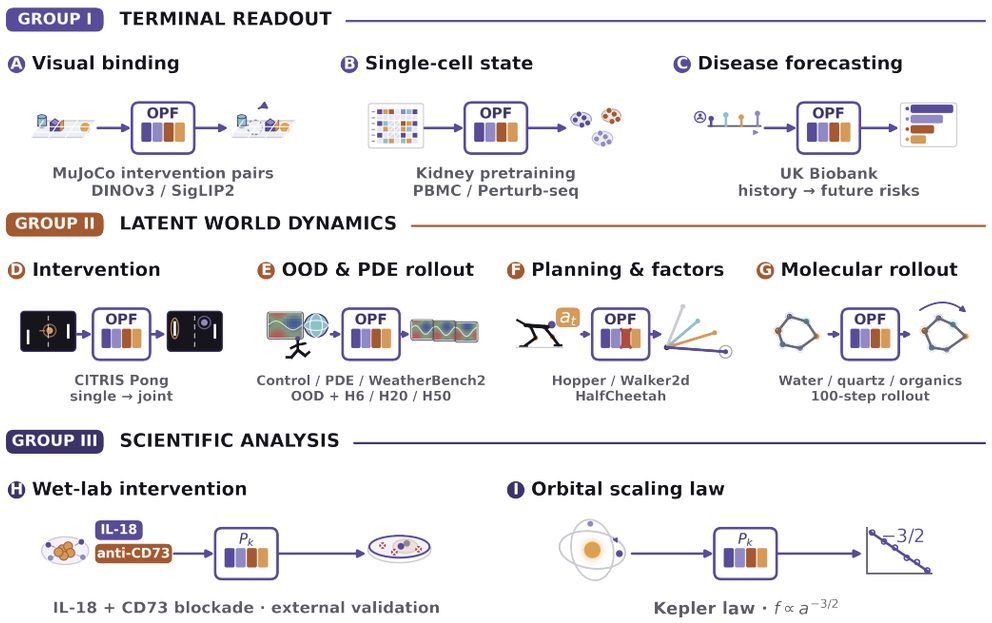


Figure 2 | Scenario atlas for JEPA-Anything. The layout follows the paper's three evaluation groups: terminal readout, latent world dynamics, and scientific analysis. Domain-specific contexts and outputs are connected through the same factorized predictive-state interface; Group III retains the learned factor coordinates for external biological validation and physical-law analysis.

1 EXTRA El atlas de escenarios sirve como mapa; no hace falta detenerse.

3.1. Common Evaluation Logic

The quantitative comparisons in Groups I and II combine domain-standard baselines, monolithic JEPA baselines, and JEPA-Anything. In direct standard JEPA comparisons, the adapter, encoder architecture, context–target sampler, optimization budget, data split, downstream readout, and inherited base loss are held fixed. The standard JEPA condition adds its monolithic predictive term, whereas JEPA-Anything adds $\mathcal{L}_{\text{OPF}}$, including the factor-specific regularizers. Other domain baselines establish reference points for the performance scale of each task. Table 2 summarizes how the shared predictive interface is instantiated and evaluated.

The quantitative tasks use their domain metrics rather than averaging raw values across incompatible scales. Continuous-control results are summarized over five training seeds and matched environment resets. Group III instead connects model-derived hypotheses or latent modes to biological experiments or established physical laws.

2 CLAVE **Control experimental:** mismo adaptador, encoder, muestreo, presupuesto, datos y readout. Solo cambia la pérdida: monolítica frente a $\mathcal{L}_{\text{OPF}}$.

3 CUIDADO "Solo cambia la pérdida", pero esa pérdida trae cuatro cosas a la vez (ecuación 10). La comparación aísla OPF **completo**, no la factorización.

4 APUNTE El grupo I termina en un readout.
El grupo II reutiliza el estado predicho en bucle, que es donde se pone a prueba un modelo del mundo.

3.2. Group I: Terminal Readout of Predictive World States

This group evaluates a predictive state at a terminal downstream readout. The visual and single-cell experiments read the online encoder state directly. The disease experiment predicts orthogonal factors, synthesizes one future patient state, and reads event risks from that state. The distinction from Group II is recursive use: Group I terminates at the readout, while Group II feeds predicted states into subsequent transition or planning steps.

1 EXTRA Tabla 2: hojéala solo para ver que K es 4 en casi todo (5 en Pong) y que las semillas van de 3 a 10.

Table 2 | Overview of JEPA-Anything instantiations and evaluation protocols. In Groups I and II, each implementation combines its base loss with $\mathcal{L}_{\text{OPF}}$ as specified in Equation (10). Numerical OPF entries use the compact form d; $K\times r$, where $Kr = d$; symbolic entries denote domain-configured widths. Group III reports the factor interface used for analysis and external validation.

Group/domain and data	Context → target or analysis	Backbone or interface	OPF setup / interface	Seeds / splits / metrics
I / vision; controlled MuJoCo scenes	Visible patches → masked patch states	DINOv3 ViT-S SigLIP2 Base	384; 4×96 768; 4×192	10 seeds; image-disjoint and leave-one-cell-out; INJ, Coll., Rec.
I / single cell; kidney, PBMC-10K, Adamson, Norman	Representation: masked expression → complete cell state; Norman: target $z_{\text{response}} := z_{\text{observed}} - z_{\text{control}}$	scGPT	Norman: 512; 4×128	5 seeds; kidney pretraining; PBMC fine-tuned/zero-shot and cross-dataset perturbation transfer; AvgBIO, Pearson
I / clinical; longitudinal multimodal cohort	Patient history → future patient state	GPT-2 small; risk decoder	768; 4×192	5 seeds; fixed patient-level cohort split; more than 1,000 event risks; PRAUC
II / intervention; CITRIS Interventional Pong	Observation + intervention → next state	Matched encoder and predictor	160; 5×32	5 paired seeds; single- and combined-intervention one-step prediction; six-step free rollout; four-channel MSE
II / dynamics; CausalWorld, DMC, PDEBench, WeatherBench2	State, frames, action, or forcing → next state or field	Matched task-specific encoder and transition	128; 4×32	5 seeds; held-out/OOD and 6-step rollout; MSE and return
II / PDE audit; APEBench Burgers and KS	Current field → next field	MLP; first-layer weight 256×64	128; 4×32	5 paired seeds; held-out late-state prediction; Burgers 6-step rollout; MSE
II / long rollout; 1-D viscous Burgers, 64 points	Current field → autoregressive future field	MLP; input $x_{\text{dim}} = 1024$; hidden width 256	128; 4×32	3 seeds; 2,000 training steps; ID/high-frequency OOD at H20/H50; MSE
II / locomotion; Hopper, Walker2d, HalfCheetah	State + action → future control state	Latent dynamics; CEM planner	32; 4×8	5 seeds; matched resets and 20-step factor interventions; return, R^2, MSE
II / molecules; water, quartz, paracetamol, benzene	Atomic state → future molecular state	TrajCast-style $O(3)$ equivariant; $l = 1$ sector	64; 4×16 (channels)	5 seeds; 100-step free rollout; MAE, RMSD
III / biology; Huh7–PBMC, organoids, tumor fragments, mice	Factor coordinates → candidate intervention nomination	OPF factor-analysis interface	Retained P_k	3 organoids and 3 fragments plus external mouse validation; killing and activation
III / orbital; simulated position–velocity trajectories	Factor coordinates → latent spectral modes	Trajectory encoding; spectral analysis	Retained P_k	One analyzed run with analytic-law validation; slope and R^2

2 PREGUNTA Pong usa $K = 5$, el único distinto. ¿Coincide con el número de variables causales conocidas del entorno? Si es así, K codifica conocimiento previo del dominio. Verificar en Lippe et al. (2022).

3 CUIDADO La validación biológica no se compara con ningún otro método para proponer intervenciones.

4 CUIDADO "One analyzed run": el resultado de Kepler viene de una sola corrida.

Table 3 | Controlled visual binding on MuJoCo, separating encoder pretraining from downstream grid supervision. Known grid supplies the support–operation taxonomy; learned grid estimates it from flat cell labels. Standard JEPA and JEPA-Anything use the same learned-grid readout.

Backbone	Encoder pretraining	Grid supervision	INJ ↑	Coll. ↓	Rec. ↑
DINOv3	Frozen checkpoint	Known grid	.452	.667	—
DINOv3	Frozen checkpoint	Learned grid	.569	.433	.643
DINOv3	Standard JEPA	Learned grid	.572	.426	.645
DINOv3	JEPA-Anything	Learned grid	.581	.417	.659
SigLIP2	Frozen checkpoint	Known grid	.476	.656	—
SigLIP2	Frozen checkpoint	Learned grid	.484	.511	.676
SigLIP2	Standard JEPA	Learned grid	.483	.514	.679
SigLIP2	JEPA-Anything	Learned grid	.490	.503	.688

1 **CUIDADO** Diferencias pequeñas: INJ .572 → .581 (DINOv3) y .483 → .490 (SigLIP2), sin desviaciones estándar aunque hay 10 semillas.

3.2.1. Controlled Visual Binding

Question. The first scenario tests whether orthogonal predictive factorization improves the compositional readout of controlled visual changes. A frozen representation must preserve both where a change occurs and which operation occurred, including support–operation combinations withheld from readout training.

2 **PREGUNTA** Esta tarea sí es composicional: **dónde** ocurre un cambio × **qué** operación, con combinaciones nunca vistas por el readout (ep. §3.4, M4). La mejora existe, pero es chica.

Predictive pretraining. For DINOv3 ViT-S ($d = 384$) [27] and SigLIP2 Base ($d = 768$) [33], the frozen baseline directly uses the original checkpoint. We use $K = 4$, giving factor widths $r = 96$ and $r = 192$, respectively. Standard JEPA and JEPA-Anything start from identical copies and are trained by block-masked patch prediction with two-dimensional position descriptors. They share the same data, masks, optimization settings, and readout architecture; standard JEPA uses a monolithic target, whereas JEPA-Anything uses the orthogonally factorized target. The resulting encoders are frozen for controlled-intervention evaluation.

3 **EXTRA** Detalles del protocolo (alineación húngara, leave-one-cell-out): se pueden saltar.

Readout and evaluation. For each source image and its controlled intervention, the patch-token difference defines an innovation field. We evaluate reconstructed MuJoCo scenes [32] with image-disjoint splits, leave-one-cell-out readout, and injective Hungarian alignment [19]. Known grid and learned grid are alternative supervision conditions for the downstream structured grid readout. Evaluating the frozen checkpoints under both separates supplied from estimated grid structure. Standard JEPA and JEPA-Anything use the same learned-grid readout, so their comparison isolates encoder pretraining. We report injective held-out-cell accuracy (INJ), collapse rate at threshold 0.5 (Coll.), and grid recovery (Rec.), averaged over ten seeds. Recovery is not applicable when the grid is supplied.

With the learned-grid readout held fixed, JEPA-Anything increases INJ and grid recovery while reducing collapse for both DINOv3 and SigLIP2.

Table 4 | Single-cell representation and perturbation evaluation. Perturbation columns report Pearson correlation on absolute post-perturbation expression.

Model	JEPA	Orthogonal factors	PBMC finetuned AvgBIO ↑	PBMC zero-shot AvgBIO ↑	Norman Pearson ↑	Adamson Pearson ↑
scGPT	No	No	0.7531	0.5288	0.631	0.905
Cell-JEPA	Yes	No	0.7830	0.7194	0.787	0.937
JEPA-Anything	Yes	Yes	0.8301	0.7752	0.814	0.942

1 CUIDADO Sin desviaciones (5 semillas). La mayor diferencia viene de **predecir en latente**: de scGPT a Cell-JEPA, 0.529 → 0.719 en zero-shot. OPF agrega menos (→ 0.775).

3.2.2. Single-Cell State Representation and Perturbation Prediction

Question. Single-cell RNA sequencing represents each cell through a high-dimensional but extremely sparse expression profile. Gene-level reconstruction can therefore devote capacity to technical dropout and count variability. This scenario asks whether latent prediction learns a more stable representation of cellular state, and whether orthogonal factorization further improves cell-type clustering and perturbation-response prediction.

2 PREGUNTA El argumento de I-JEPA en otro dominio: reconstruir genes gasta capacidad en ruido técnico (dropout), y predecir en latente lo evita (M1, ep. §3.1).

Protocol. Following Cell-JEPA [10], each cell is tokenized by gene identity and discretized expression value using an scGPT backbone [9]. The representation-learning path follows the masked Cell-JEPA construction: a student encoder receives masked expression values and predicts the cell-level embedding produced by an EMA teacher from the unmasked cell. The Norman perturbation experiment predicts the latent response $z_{\text{response}} = z_{\text{observed}} - z_{\text{control}}$. Its 512-dimensional response representation uses $K = 4$ factors of width $r = 128$. Cell-JEPA combines its monolithic latent loss with masked-gene reconstruction. We re-evaluate it as the monolithic JEPA comparison in Table 4, while JEPA-Anything applies factorized latent prediction to the target used by each experiment. Perturbation evaluation follows the encoder → factor prediction heads → gating → latent response residual → expression decoder path.

Evaluation. Models are pretrained on approximately 800,000 human kidney cells and evaluated on PBMC-10K cell-type clustering under finetuned and zero-shot settings. We also evaluate perturbation-response prediction on the Adamson [1] and Norman [24] datasets. AvgBIO summarizes clustering quality, while Pearson correlation measures absolute post-perturbation expression prediction. All reported results are averaged over five training seeds.

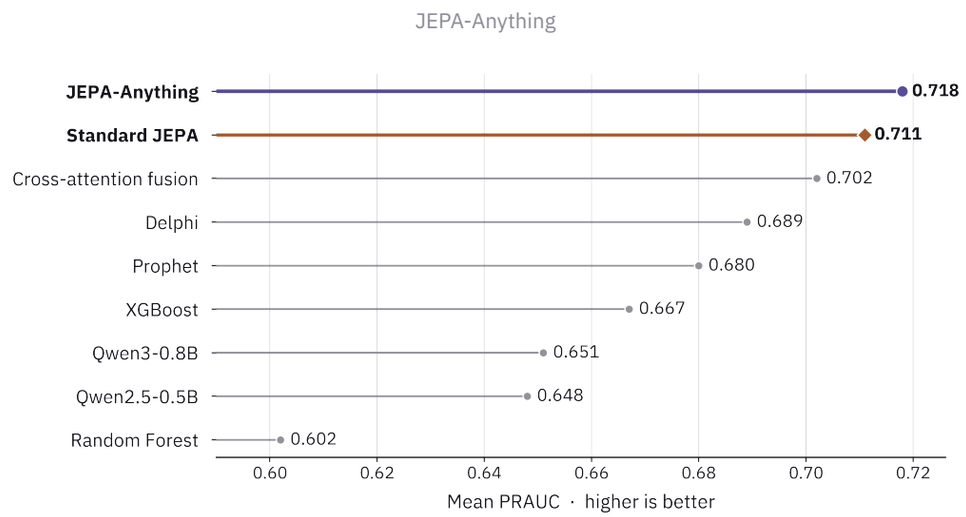
JEPA-Anything attains the highest value on both PBMC clustering metrics and both perturbation-response datasets.

3 EXTRA Dominio clínico: lectura rápida. Mira solo la conclusión.

3.2.3. Longitudinal Disease-State Forecasting

Question. Longitudinal health records are sparse, irregularly timed, and only partially observe the physiological state that drives disease progression. This scenario tests whether predicting future latent health states supports broad and long-range clinical-event forecasting, and whether orthogonal factorization improves prediction across the full event vocabulary.

Protocol. Each patient state combines time-ordered clinical events, demographic context, and molecular or biochemical variables. The context encoder and its EMA target copy



1 EXTRA Figura 3: ranking por PRAUC medio. No muestra el tamaño de la diferencia con el JEPA monolítico ni su variabilidad.

Figure 3 | Broad-spectrum prediction of more than 1,000 future clinical events. Methods are ranked by mean PRAUC using the same cohort split and event vocabulary; higher is better.

use the GPT-2 small architecture with hidden width $d = 768$ [25]; the former represents past history and the latter a future patient state. We use $K = 4$ factors of width $r = 192$. A monolithic health-state JEPA provides the direct comparison. The JEPA-Anything condition predicts orthogonal factors and reconstructs the complete future state with Equation (5). A shared lightweight decoder maps the resulting state to risks for more than 1,000 future clinical events. Figure 3 compares the resulting model rankings.

Evaluation. Mean area under the precision–recall curve (PRAUC) across the full event vocabulary is the primary metric because many outcomes are rare. All methods use five training seeds and the same fixed patient-level cohort split.

The factorized future state yields a higher mean PRAUC than the matched monolithic JEPA across the event vocabulary.

3.3. Group II: Latent World Dynamics and Rollout Stability

2

This group tests the world-modeling role of the factorized state directly: predicted latent states are used repeatedly by a planner or fed back through an autoregressive rollout, so representation errors become dynamical errors. It therefore evaluates the operational world-model mode. It includes controlled and visual dynamics, physical fields, weather forecasting, closed-loop planning, and molecular trajectories.

2 CLAVE **El grupo II es el corazón para P1:** el estado predicho se reutiliza en bucle, así que los errores de representación se vuelven errores **dinámicos** (M2).

3.3.1. Intervention-Conditioned Compositional Prediction

3

Question. CITRIS Interventional Pong [21] is a controlled dynamical environment based on a simplified Pong game. Each sequence records how the game state evolves and also identifies which underlying state factors were externally changed between observations. Here, an intervention is not an action chosen by the model: the experiment changes one or more factors of the system, and the model receives both the pre-intervention observation and a label specifying what was changed. Its task is to predict the resulting next state. Figure 4 summarizes the paired prediction results.

3 CLAVE **CITRIS Pong:** se interviene una o varias variables del juego y el modelo recibe la etiqueta de qué cambió. Debe predecir el estado siguiente.

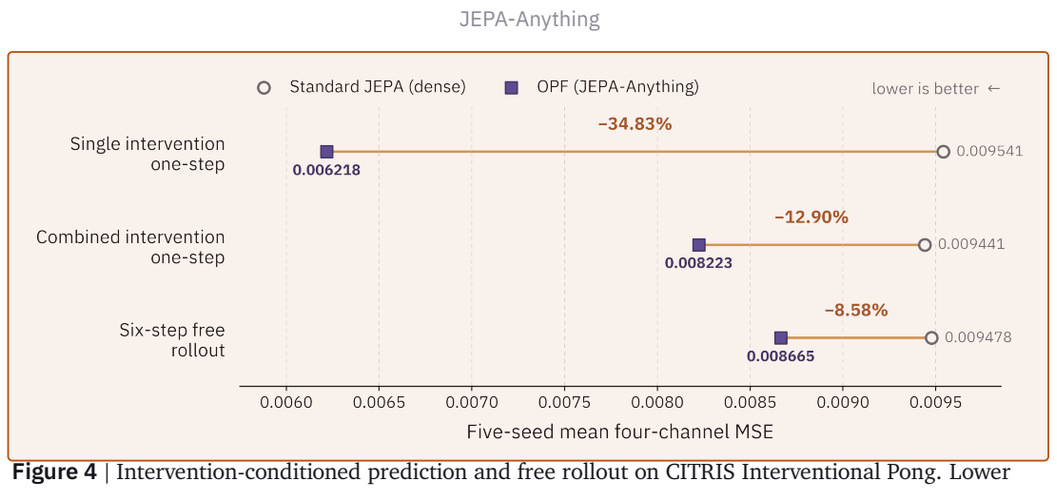


Figure 4 | Intervention-conditioned prediction and free rollout on CITRIS Interventional Pong. Lower four-channel MSE is better. Points show means over five paired seeds; annotations give the reduction computed from the reported means.

1 **CUIDADO** **Mira los dos primeros renglones.** El JEPA estándar tiene casi el mismo error en intervenciones simples y combinadas (0.00954 y 0.00944). JEPA-Anything mejora mucho las simples (0.00622) y menos las combinadas (0.00822): **la brecha composicional crece con OPF**.

We evaluate two levels of generalization. In the first, a single factor is intervened on in a way represented in the training distribution. In the second, multiple factors are intervened on together in a combination that was withheld from training, even though the individual types of change were observed separately. The latter condition therefore asks whether the model merely memorizes complete intervention patterns or can recombine learned state-change rules when they occur together in a new configuration. Lower prediction error in both settings indicates more accurate intervention-conditioned dynamics and better compositional reuse of previously learned changes. We additionally evaluate six-step free rollout.

2 **CLAVE** Dos niveles de generalización: (1) una intervención vista; (2) una **combinación nunca vista** de intervenciones vistas por separado. El segundo es generalización composicional en sentido estricto (ep. §3.4; Lake y Baroni).

3 **PREGUNTA** "Recombinar reglas de cambio aprendidas": si cada factor capturara una variable, intervenir dos variables sería cambiar dos factores. Es la promesa de "factor = regla"; revisa si la evidencia alcanza (C5).

Protocol and evaluation. The dense standard JEPA baseline and JEPA-Anything use the same data, model budget, and five fixed training seeds. We report the mean four-channel MSE for single-intervention and combined-intervention one-step prediction, and for six-step free rollout.

Relative to the dense standard JEPA baseline, JEPA-Anything reduces single-intervention one-step MSE from 0.009541 to 0.006218, combined-intervention one-step MSE from 0.009441 to 0.008223, and six-step free-rollout MSE from 0.009478 to 0.008665. The corresponding reductions are 34.83%, 12.90%, and 8.58%, respectively.

4 **APUNTE** Auditoría: misma capacidad, pero con cabezas sin restricción de ortogonalidad. Ojo: solo mide **geometría**, no error de predicción.

Orthogonality mechanism audit. We separately isolate the structural role of orthogonality by comparing the factor interface with a capacity-matched unconstrained multi-head condition on the same official CITRIS data, five training seeds, and 2,500-step budget. The audit measures whether the learned factor blocks overlap, whether their combined analysis matrix covers the state space with stable singular values, whether the original latent state can be synthesized directly from the factor coordinates, and whether this structure is obtained without deactivating most factors.

Table 5 | Mechanism audit of the factor interface on CITRIS Interventional Pong. Entries are mean ± sample standard deviation over five paired seeds.

Metric	Unconstrained multi-head	Orthogonal factorization
Cross-factor subspace overlap ↓	0.4550 ± 0.0420	$(5.18 \pm 0.34) \times 10^{-16}$
Minimum singular value $\sigma_{\min}(P)$ ↑	0.00513 ± 0.00233	0.999989 ± 0.000003
Condition number $\kappa_2(P)$ ↓	438.52 ± 170.00	1.00005 ± 0.000004
Transpose synthesis NMSE (exact factors) ↓	0.7886 ± 0.0299	$(2.98 \pm 0.11) \times 10^{-14}$

1 CLAVE **Tabla 5.** Con OPF, el solapamiento es casi 0, $\sigma_{\min} \approx 1$, $\kappa_2 \approx 1$ y la reconstrucción es exacta. Sin restricción, $\kappa_2 \approx 439$. Es la proposición 1 confirmada en un modelo entrenado.

The four metrics isolate complementary geometric properties. Cross-factor overlap measures redundancy between factor subspaces; $\sigma_{\min}(P)$ measures the weakest-covered state direction; and $\kappa_2(P)$ measures the sensitivity of the coordinate system to errors. The reported synthesis NMSE uses exact coordinates $u = P^\top z$ and transpose synthesis Pu, providing a measure of factor-interface geometry independent of predictor error. Together, the results show that orthogonal factorization yields a self-dual, unit-conditioned coordinate system in which P synthesizes the state from $P^\top z$. Orthogonality therefore turns multiple prediction outputs into a complete, non-overlapping, and numerically stable world-state interface.

2 CUIDADO Pero es casi una tautología: si se **penaliza** la falta de ortogonalidad, el resultado es ortogonal. Falta lo importante: ¿predicen peor las cabezas sin restricción, en un paso y en rollout? No se reporta.

3.3.2. Out-of-Distribution and Long-Horizon Dynamics

Question. This benchmark evaluates two capabilities required of a predictive world model. First, after learning from a limited set of trajectories, the model must predict states from unseen initial conditions, held-out episodes, or later time periods. This out-of-distribution test asks whether the model has learned a reusable transition rule rather than memorized the trajectories used for training. Second, the model is rolled forward for six consecutive steps. At each step its prediction becomes part of the input for the next prediction, so small errors can compound over time. A model with more stable learned dynamics should therefore maintain lower error as the rollout proceeds.

3 CLAVE Dos pruebas de modelo del mundo: fuera de distribución (¿aprendió una regla de transición reutilizable o memorizó?) y rollout de 6 pasos (¿se acumula el error?).

4 PREGUNTA Aquí debería verse el mecanismo M2. Para verlo, compara cuánto **crece** el error del paso 1 al 6, no solo su nivel (tabla 6, pág. 17).

We apply the same evaluation principle to systems represented as moving pixels, robot states and control-conditioned observations, and continuous physical or weather fields. These settings are instantiated with CausalWorld [2], the DeepMind Control Suite [30], PDEBench [29], and WeatherBench2 [26]. The comparison tests whether assigning complementary parts of a latent state transition to separate predictive factors produces more accurate predictions on unseen states and slows error growth during repeated prediction.

5 EXTRA Protocolo: 10 tareas, 5 semillas, mismo backbone y presupuesto.

Protocol. The benchmark compares standard JEPA and JEPA-Anything over ten tasks and five random seeds $\{11, 23, 37, 53, 71\}$. Standard JEPA uses one shared latent target, whereas JEPA-Anything decomposes the target into four factors. Within each task and seed, both variants use the same data, encoder and transition backbone, optimization budget, evaluation split, and compute backend.

All rollout tasks receive the true contiguous six-step action or target sequence, predict one step at a time, and compare each prediction with ground truth at the corresponding time. WeatherBench2 normalization uses statistics from the training period only, and

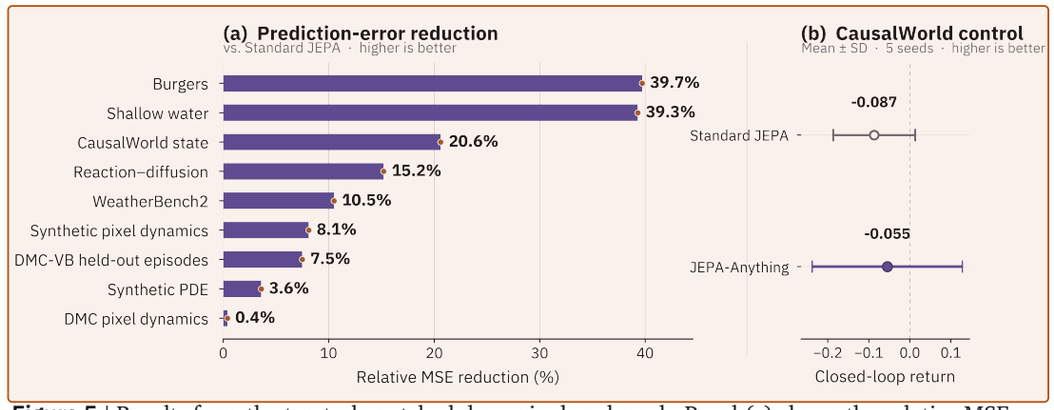


Figure 5 | Results from the ten-task matched dynamics benchmark. Panel (a) shows the relative MSE reduction of JEPA-Anything versus standard JEPA, computed from the reported five-seed means for the nine prediction tasks. Panel (b) shows CausalWorld closed-loop control return as mean ± sample standard deviation over five seeds; higher return is better.

control inputs use post-intervention observations. Figure 5 reports relative changes computed from the five-seed means for the prediction tasks and mean ± sample standard deviation for CausalWorld control.

Rollout and factor diagnostics. Table 6 reports the first and sixth rollout steps for four representative systems, making the accumulation of prediction error explicit.

Table 6 | Six-step rollout diagnostics. Lower MSE is better. Each entry is averaged over five seeds; the full evaluation records every intermediate step.

	Standard JEPA		JEPA-Anything	
Task	Step 1	Step 6	Step 1	Step 6
CausalWorld state prediction	0.01306	0.01307	0.01037	0.01079
DMC pixel dynamics	0.008539	0.008594	0.008498	0.008526
PDEBench Burgers	0.001830	0.006369	0.001101	0.004014
PDEBench shallow water	0.007090	0.010510	0.003999	0.006522

Additional APEBench evaluation. We additionally evaluate Burgers and Kuramoto–Sivashinsky (KS) dynamics under the autoregressive PDE protocol of APEBench [18]. We report these results separately because APEBench and PDEBench use different normalizations. The APEBench models use an MLP with a first linear-layer weight of shape $[256, 64]$ and OPF dimensions $d = 128$, $K = 4$, and $r = 32$. Table 7 reports held-out late-state prediction for both systems and the six-step Burgers rollout.

Table 7 | Prediction results in the additional APEBench evaluation. Lower MSE is better.

System	Metric	Standard JEPA	JEPA-Anything
Burgers	Held-out late MSE ↓	0.16621 ± 0.01740	0.08389 ± 0.01770
Burgers	Six-step rollout MSE ↓	0.29186 ± 0.00682	0.16153 ± 0.03500
Kuramoto–Sivashinsky	Held-out late MSE ↓	1.1012 ± 0.0763	0.9558 ± 0.0340

1 **CUIDADO** Figura 5a: de 39.7 % (Burgers) a 0.4 % (DMC píxeles); cuatro de las nueve tareas mejoran menos de 10 %. Figura 5b: en control, ambos retornos son negativos y los intervalos se solapan, así que no hay mejora clara.

2 **PREGUNTA** OPF no "redondea" estados como un latente discreto (los categóricos de DreamerV3). Un experimento interesante: el mismo rollout con latentes categóricos (M2, P1).

3 **CUIDADO** **Revisa el crecimiento del error.** Burgers: ×3.48 (estándar) frente a ×3.65 (OPF). Aguas someras: ×1.48 frente a ×1.63. CausalWorld: ×1.00 frente a ×1.04. OPF parte con menos error, pero el error **crece igual o más rápido**: la ventaja es de precisión a un paso, no de estabilidad.

4 **EJERCICIO** Calcula las razones paso 6 / paso 1 de cada fila. ¿Qué esperarías ver si OPF realmente frenara la acumulación de error?

5 **CLAVE** APEBench: mejoras grandes y consistentes (Burgers, 45–50 %; KS, 13 %), con desviaciones reportadas y en todas las semillas. Es de los resultados más sólidos del paper.

The Burgers errors decrease by approximately 49.5% on the held-out late segment and 44.7% over six rollout steps, with improvement in every seed. KS late-state prediction improves by approximately 13.2%, also in every seed. This extends the improvement to chaotic nonlinear dynamics.

Capacity-matched Burgers long-rollout evaluation. We additionally evaluate a parameter-matched 50-step protocol. Standard JEPA and JEPA-Anything use the same 1-D periodic viscous Burgers data with 64 grid points and $\nu = 0.01$, and are evaluated over three seeds on both in-distribution and high-frequency out-of-distribution trajectories. The emulator receives an input of dimension $x_{\text{dim}} = 1024$ and uses an MLP with hidden width 256. Its latent state has $d = 128$ dimensions and is divided into $K = 4$ factors of width $r = 32$, giving a stacked projector shape of $(4, 128, 32)$. Models are trained for 2,000 steps. Table 8 reports the 20- and 50-step errors.

Table 8 | Capacity-matched Burgers long-rollout evaluation over three seeds. Lower error is better. Relative change is computed against standard JEPA.

Split	Standard H20 / H50	JEPA-Anything H20 / H50	JEPA-Anything relative change
ID	1.1185 / 0.8181	1.0532 / 0.7923	−5.84% / −3.15%
High-frequency OOD	1.0649 / 0.8452	1.0116 / 0.8214	−5.01% / −2.82%

The gain persists on both splits and at both horizons, including high-frequency OOD trajectories, although it narrows from H20 to H50. The evaluation extends the shorter APEBench rollout results to 50 prediction steps.

For the nine predictive tasks with routing statistics, JEPA-Anything uses an average of 3.782 effective factors out of four, with an average dominant-factor share of 0.324, indicating distributed activity across the four branches.

3.3.3. Continuous-Control Planning

Question. In model-based control, the agent uses the learned world model as an internal simulator. Figure 6 reports the comparison between standard JEPA and the full JEPA-Anything model. At each decision point, the CEM planner generates many candidate action sequences, rolls each sequence forward through the model, estimates its future return, and executes the first action from the highest-scoring sequence. The model must therefore predict not only the immediate next state but a chain of states under different possible actions. Even a small latent prediction error can accumulate across this imagined trajectory, causing the planner to rank a poor action sequence above a genuinely better one.

This experiment asks whether organizing the predicted state into orthogonal factors improves the quality of these planning decisions. Standard JEPA and JEPA-Anything are matched in parameter count and prediction cost, isolating the effect of factorization. The final evaluation uses the return obtained in the real environment because the central question is whether the learned latent dynamics lead the planner to choose better actions.

1 CUIDADO La ventaja se **reduce** con el horizonte: −5.8 % a H20 y −3.2 % a H50. Además, el error a H50 es menor que a H20 en ambos modelos; quizá porque Burgers viscoso se disipa hacia un estado suave (hipótesis: verifícala).

2 PREGUNTA 3.78 factores efectivos de 4, sin uno dominante (0.32). Es la única medida de **cuántos** factores usa el modelo. Si con $K = 8$ siguieran todos activos, ¿hay 8 "cosas" que predecir o es solo capacidad repartida? (C4).

3 CLAVE Planificación con CEM: se imaginan muchas secuencias de acciones y se ejecuta la primera de la mejor. Un error latente pequeño puede invertir el orden de las acciones.

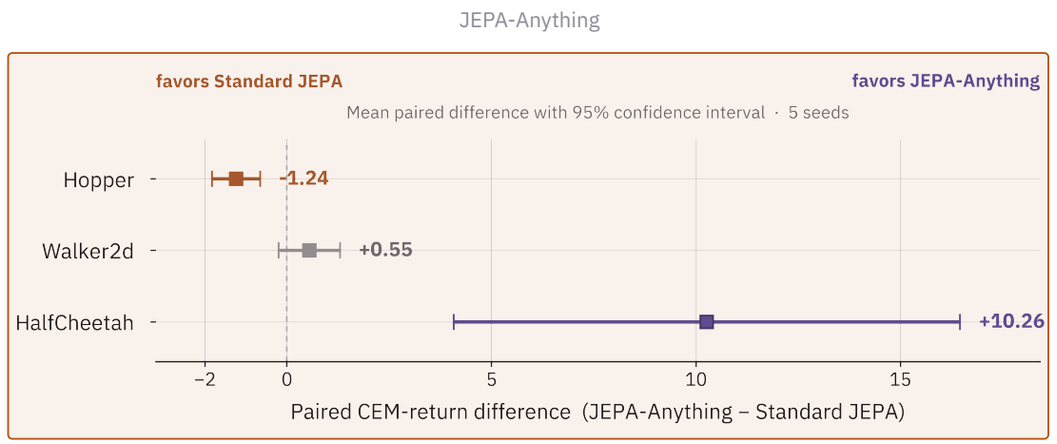


Figure 6 | Capacity-matched continuous-control planning. Squares show the mean paired CEM-return difference between JEPA-Anything and standard JEPA over five training seeds; error bars show 95% confidence intervals. Positive values favor JEPA-Anything.

Protocol. We evaluate Hopper-v5, Walker2d-v5, and HalfCheetah-v5 [32]. A dense, capacity-matched standard JEPA and the full JEPA-Anything model with $K = 4$ use the same data, backbone, optimization budget, and evaluation protocol. Training uses seeds $\{42, 123, 456, 789, 2026\}$. Each checkpoint is evaluated on five identical environment resets, and paired comparisons reset the CEM candidate-action randomness.

Branch widths are calibrated for $K \in \{1, 2, 4, 8\}$. Relative to dense standard JEPA, the training-parameter differences are 0.000%, 0.006%, 0.126%, and 0.246%, while prediction-head FLOP differences are 0.000%, 0.742%, 0.992%, and 1.449%, respectively, all below 0.3% in parameters and 2% in FLOPs.

JEPA-Anything attains higher mean CEM return on Walker2d and HalfCheetah, while Hopper favors standard JEPA. Planning performance is therefore environment-dependent. The factor-level interventions below test whether every learned factor is functionally used during rollout and planning.

3.3.4. Factor Activity and Functional Interventions in Locomotion

Question and protocol. This experiment tests whether the four orthogonal groups in the locomotion world model form active and individually usable state channels. For the full $K = 4$ model, the 32-dimensional state is partitioned into four learned eight-dimensional subspaces. On held-out transitions, a separate ridge probe uses one factor at a time to predict the next-step observation change $\Delta o_t = o_{t+1} - o_t$. We additionally measure factor-wise standard deviation, replace one factor by its training-set mean throughout a 20-step rollout, and repeat the intervention during paired CEM evaluation in the real environment.

Descriptive readout. The single-factor probes reveal differentiated motion preferences. On Hopper, the most readable changes from F1–F4 are foot angle ($R^2 = 0.294$), torso height (0.347), torso angle (0.424), and leg angle (0.310), respectively. These are linearly accessible motion preferences rather than preassigned one-to-one factor labels; one factor may retain information about several related variables.

1 **CUIDADO** Figura 6. En Hopper gana el JEPA estándar, con un intervalo que no cruza el 0. En Walker2d el intervalo cruza el 0 (no hay diferencia). Solo HalfCheetah favorece a OPF, con un intervalo muy ancho: 1 de 3.

2 **CUIDADO** **Calibran anchos para $K \in \{1, 2, 4, 8\}$, pero no reportan resultados para esos K.** Es exactamente el experimento que pide C4 (¿cuántos factores?), y no está.

3 **PROFUNDIZAR** Experimento que valdría la pena: barrer K y elegirlo con un criterio (MDL, verosimilitud marginal, ganancia predictiva marginal). Sería OPF "a lo DreamCoder": ampliar solo si la nueva pieza paga su costo (ep. §4.4).

4 **CLAVE** Prueba en tres partes: ¿cada factor está activo? (varianza), ¿es legible? (sonda lineal) y ¿se usa? (reemplazarlo por su media y ver cuánto empeora).

5 **PREGUNTA** R^2 de 0.29 a 0.42: cada factor se **inclina** hacia una variable (pie, torso…) pero no la captura. No es discreto ni se puede nombrar con nitidez: un factor **no** es todavía una regla (C5).

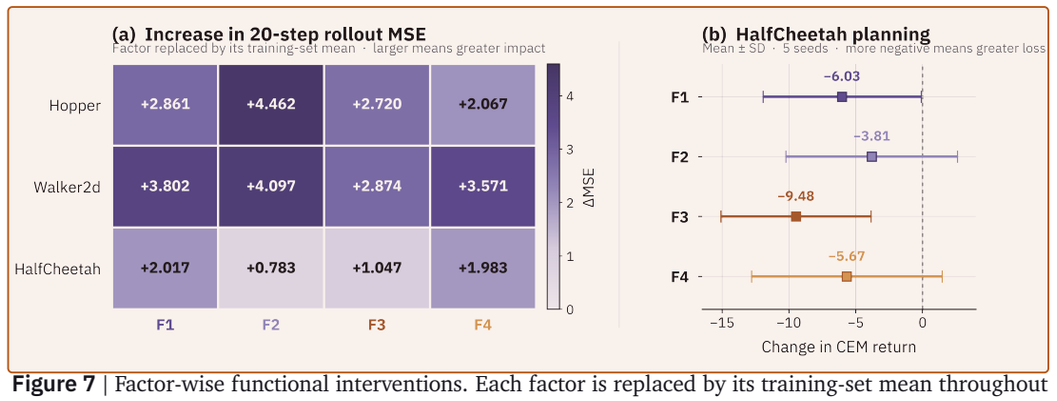


Figure 7 | Factor-wise functional interventions. Each factor is replaced by its training-set mean throughout evaluation. Panel (a) gives the increase in 20-step rollout MSE. Panel (b) gives the HalfCheetah CEM-return change as mean ± sample standard deviation over five training seeds.

Activity and intervention. All four groups remain active. Across the analyzed checkpoints, their factor-wise standard deviations lie in 0.692–0.715 on Hopper, 0.587–0.601 on Walker2d, and 0.673–0.678 on HalfCheetah. Figure 7 reports the change caused by masking one factor. Positive heatmap values indicate increased 20-step MSE; negative CEM-return changes indicate reduced real-environment return.

Replacing a factor by its training-set mean removes sample-specific information while keeping the intervened coordinate in a typical numerical range. Masking any factor increases rollout error in all three environments. In HalfCheetah, masking F1 or F3 reduces return in all five training seeds, while masking F2 or F4 reduces return in four of five.

Together, factor activity, single-factor readout, and masking form an active–readable–functionally-used evidence chain. They show that orthogonal predictive factorization supplies geometrically non-overlapping, active, and individually intervenable state channels.

3.3.5. Force-Free Molecular Forecasting

Question. Force-free molecular forecasters directly predict future atomic positions and velocities, then feed their own outputs back autoregressively. Small local errors can therefore accumulate even when one-step prediction is accurate. This scenario tests whether orthogonal future factors improve one-step accuracy and long-horizon molecular forecasting stability.

Protocol. Each molecular state contains atomic positions, velocities, species, and the simulation cell. We use a TrajCast-style equivariant forecaster [31] on liquid water, crystalline α-quartz, gas-phase paracetamol, and benzene, and denote the JEPA-pretrained reference condition as TrajCast-JEPA. JEPA-Anything retains the same system data, $O(3)$-equivariant backbone, prediction intervals, supervised adaptation, and rollout protocol while applying orthogonal predictive factorization during pretraining. We apply OPF to the 64-channel multiplicity axis of the $l = 1$ sector, using $d = 64$, $K = 4$, and $r = 16$. Because an $l = 1$ irrep has three components, the corresponding flattened block contains $64 \times 3 = 192$ scalar coordinates; the learned channel projections act identically on the three components, equivalently as $P_k \otimes I_3$, to preserve equivariance.

1 **CUIDADO** Falta una línea de base: borrar 8 de 32 dimensiones de **cualquier** estado aprendido empeoraría el rollout. Habría que hacer lo mismo con 8 direcciones al azar, o con un subespacio del JEPA estándar.

2 **CLAVE** La cadena activo → legible → usado muestra que los 4 canales no son decorativos, pero no que correspondan a "partes del mundo".

3 **EXTRA** Moléculas: lectura rápida. Detalle técnico: OPF se aplica respetando la equivariancia $O(3)$, con $P_k \otimes I_3$.

Table 9 | Force-free molecular forecasting with a TrajCast-style backbone. Entries are means over five trained seeds. For each seed, we compute one-step displacement MAE and median final-position RMSD after 100 autoregressive steps. Values are reported in Å; lower is better.

System	Metric ↓	Scratch	TrajCast-JEPA	JEPA-Anything
Water	MAE	0.00387	0.00452	**0.00376**
Water	RMSD	3.331	2.536	**2.459**
Quartz	MAE	0.01080	0.01043	**0.01011**
Quartz	RMSD	2.089	1.912	**1.877**
Paracetamol	MAE	0.00901	0.00777	**0.00705**
Paracetamol	RMSD	3.155	1.868	**1.776**
Benzene	MAE	2.59×10^{-5}	2.10×10^{-5}	$\mathbf{2.05\times10^{-5}}$
Benzene	RMSD	0.0958	0.0701	**0.0645**

1 **CUIDADO** Mejoras consistentes pero pequeñas frente a TrajCast-JEPA (agua: RMSD 2.536 → 2.459), sin desviaciones. La mayor ganancia vuelve a ser **pasar de cero a JEPA** (3.331 → 2.536).

Evaluation. We report means over five independently trained models. For each seed, we compute the one-step displacement MAE and the median final-position RMSD after 100 free autoregressive steps, and then average each metric across the five seeds. The latter is the primary measure of rollout stability; lower values are better. Table 9 reports the values in Å.

JEPA-Anything produces the lowest one-step MAE and 100-step RMSD on all four molecular systems.

3.4. Group III: Scientific Analysis of Learned World States

This group tests factor coordinates as a scientific interface. The cancer case uses them to nominate an intervention evaluated experimentally, while the orbital case tests extracted modes against an established scaling law. Both connect factor-level analysis of $u_k(x)$ to external validation.

2 **PREGUNTA** Grupo III: los factores como "generador de hipótesis". Es lo más cercano a la abducción de Peirce en el paper (ep. §2.2, C6).

3.4.1. Wet-Lab Evaluation of a Factor-Nominated Intervention

Domain-specific intervention analysis applied to the orthogonal factor coordinates learned by JEPA-Anything nominated IL-18 combined with *NT5E*/CD73 blockade as a candidate intervention. Wet-lab studies subsequently supported this prediction, showing enhanced antitumor activity in Huh7–PBMC co-culture, patient-derived organoids and tumor fragments, and immunocompetent mice. This result illustrates how orthogonal factors learned by JEPA-Anything can support the formulation of biologically testable hypotheses.

3 **CUIDADO** **No se describe cómo** el análisis de factores nominó IL-18 + CD73, ni cuántos candidatos se evaluaron, ni si otro método (o la literatura) lo habría propuesto. Sin eso, no se puede atribuir el hallazgo a los factores.

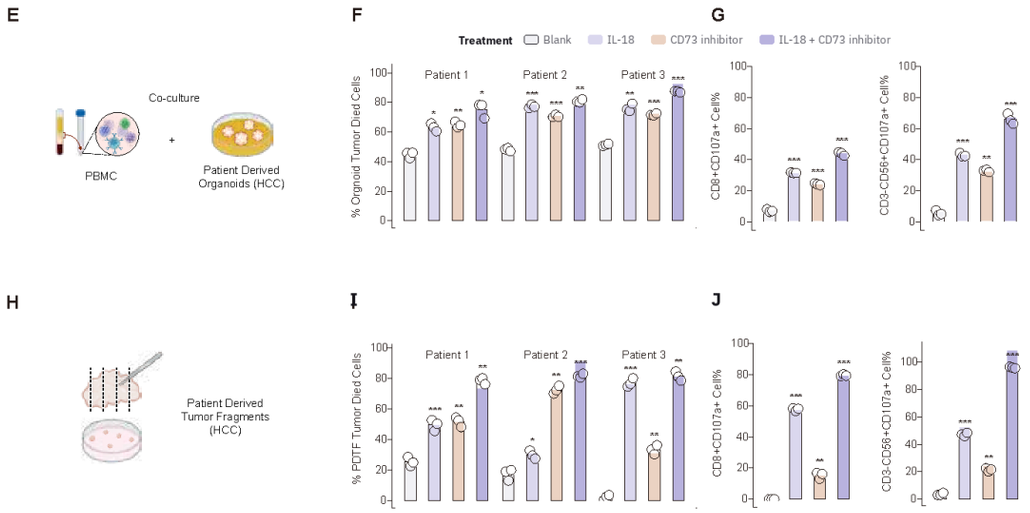


Figure 8 | Selected wet-lab results in patient-derived hepatocellular carcinoma organoids and tumor fragments. Across three organoid specimens and three tumor-fragment specimens, the IL-18 plus CD73 blockade combination showed the strongest tumor-cell killing among the tested conditions, together with increased immune-cell activation. Colors identify blank control, IL-18, CD73 inhibition, and the combined intervention.

1 **EXTRA** Resultados de laboratorio: no hace falta evaluarlos. Lo relevante es la nota anterior sobre el método de nominación.

3.4.2. Orbital Scaling as a Physical-Law Diagnostic

Astronomical trajectories provide a complementary example in which predictive latent modes are inspected for an interpretable law rather than used only for forecasting. From simulated position–velocity trajectories without physical labels, spectral analysis pairs latent frequencies with orbital semimajor axes. In the JEPA-Anything analysis shown in Figure 9, the resulting modes recover the Keplerian relation $f \propto a^{-3/2}$ with high precision.

2 **PREGUNTA** ¿Descubrir o reconocer? La ley de Kepler ya se conocía: el análisis **comprueba** que una ley sabida aparece en el latente. Es validación, no descubrimiento. Aun así, muestra que los factores capturan algo físico (C5, C6).

This example illustrates the scientific role of factor-level analysis: learned predictive components can be evaluated against candidate symmetries, invariants, or scaling relations. Here, agreement with the established Keplerian law tests whether the orthogonal factors learned by JEPA-Anything capture a physically meaningful dynamical regularity.

3 **CUIDADO** Falta el control: ¿un JEPA monolítico, o un análisis espectral de las trayectorias crudas, también recupera la pendiente −3/2? La frecuencia orbital ya está en la periodicidad de los datos.

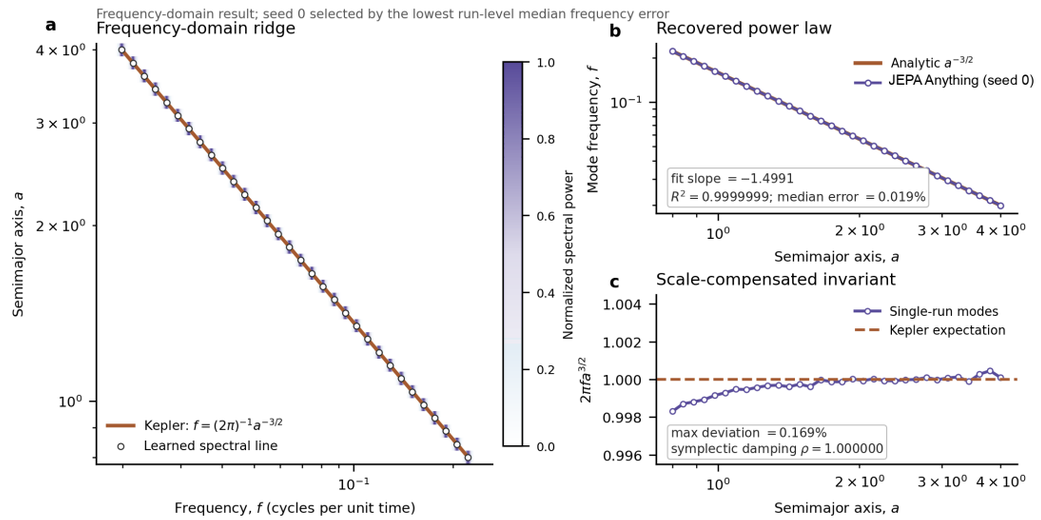


Figure 9 | A physical-law diagnostic based on predictive latent modes. In a JEPA-Anything run, the learned spectral modes recover the Keplerian ridge $f = (2\pi)^{-1}a^{-3/2}$ with fitted slope -1.4991 and $R^2 = 0.9999999$.

4. Related Work

1 **PROFUNDIZAR** Línea de modelos del mundo: Dreamer (ya visto), TD-MPC2, DINO-WM y V-JEPA 2. Para seguir de JEPA a planificación: V-JEPA 2 (ref. 4).

Latent world models and planning. Latent world models compress observations into states that can be advanced by learned dynamics. Early recurrent world models coupled a latent dynamics model to a controller [12], while DreamerV3 demonstrated that a single model-based reinforcement-learning configuration can support behavior learning across diverse domains [14]. Decoder-free approaches such as TD-MPC2 optimize latent dynamics directly for scalable continuous control [15]. DINO-WM instead predicts features from a pretrained visual encoder and plans toward visual goals from offline trajectories [37]. V-JEPA 2 combines large-scale representation learning with an action-conditioned latent model for robotic planning [4]. These systems primarily develop latent states for visual prediction or control.

2 **PROFUNDIZAR** Ref. 28, Sobal et al. (2022): los JEPA se enfocan en **rasgos lentos**, lo que cambia poco entre contexto y objetivo. Útil para entender qué decide no explicar un JEPA.

Joint-embedding predictive learning. I-JEPA predicts representations of masked image regions from visible context [3], and V-JEPA extends feature prediction to video without pixel reconstruction [7]. Subsequent work scales this principle to physical understanding and planning [4], while Cell-JEPA adapts latent prediction to single-cell transcriptomics [10]. Analyses of JEPA objectives also show that their learned invariances depend on which signals vary across context–target pairs [28].

4 **PROFUNDIZAR** Ref. 17, Kipf et al. (2020): modelos del mundo con objetos y relaciones explícitas. Buen contraste para P1: estructura discreta explícita frente a factores continuos.

Structured and factorized representations. Factorized representation learning has commonly sought coordinates aligned with independent generative variables, as in β-VAE [16], or entity-aligned slots, as in Slot Attention [23]. Structured world models similarly represent states as interacting objects and relations [17], while CITRIS uses temporal interventions to identify causal factors [21]. Such semantic decompositions require corresponding inductive biases or supervision; in general, unsupervised semantic disentanglement is not identifiable from observations alone [22].

3 **CLAVE** La diferencia con β-VAE, Slot Attention o CITRIS: esos buscan factores **semánticos** (variables generativas, objetos, causas). OPF no promete semántica; solo reparte capacidad.

5 **CUIDADO** Locatello et al. (2019): sin sesgos inductivos ni supervisión, el **disentanglement** no supervisado no es identificable. Por eso los factores de OPF no tienen por qué coincidir con variables reales (ep. Guía B).

Redundancy reduction and collapse prevention. Barlow Twins reduces redundancy through cross-correlation matching [36], and VICReg combines invariance with explicit

variance and covariance regularization [6]. Across these lines, OPF factorizes predictive target capacity without assuming object, causal, or semantic factor identities. It applies orthogonality and activity constraints to learned target-subspace bases and retains an explicit synthesis map, linking redundancy reduction to a complete latent world state.

1 CLAVE Posicionamiento honesto: OPF reparte capacidad "sin suponer identidades de objeto, causa o semántica". Tenlo presente al leer las secciones "científicas".

5. Summary

JEPA-Anything separates domain-specific predictive-state construction from a shared OPF core. OPF assigns target capacity to complementary subspaces, keeps them active, and synthesizes their predictions into a complete latent state. The same latent world-state interface supports terminal encoder readout, recursive transition and planning, or factor-level analysis.

Experiments evaluate these three uses across visual, biological, clinical, control, molecular, physical-field, and weather systems. Terminal-readout tasks report stronger visual, cellular, and disease-state metrics; CITRIS and the dynamics benchmarks support intervention composition, OOD prediction, and repeated rollout. The geometry analysis shows non-overlapping, unit-conditioned coordinates and near-exact synthesis. Continuous-control means improve on two environments but not Hopper, while factor interventions show that all locomotion channels contribute to rollout. Biological and orbital cases further illustrate external analysis of the learned coordinates. Together, the results support OPF as a capacity-organizing core for a reusable latent world-state interface.

2 CUIDADO El resumen admite lo de Hopper, pero repite que el método respalda los "rollouts repetidos", algo que la tabla 6 no respalda del todo (el error crece igual).

6. Future Work

JEPA-Anything provides a common predictive learning principle across different worlds. A broader direction is to turn these predictive models into components of systems that actively investigate the worlds they model. This connects naturally to Discovery Foundation Models (DFMs) [35], which frame discovery as a process of formulating problems, constructing representations, proposing hypotheses, and revising them through external evidence. Within this perspective, JEPA-Anything could provide a predictive substrate for comparing candidate interventions and identifying discrepancies between expected and observed outcomes. Such discrepancies could guide new experiments and, when necessary, motivate revisions to the variables, representations, or hypotheses used to describe a system.

3 PREGUNTA Las discrepancias entre lo esperado y lo observado deberían "motivar revisiones de las variables o hipótesis": es el ciclo conjeturar–refinar de P2 (ep. §4.4), pero solo como plan futuro.

4 EXTRA ScienceIDE, ScienceBuddy y DFM son proyectos de los mismos autores (PhAI Labs). Se puede saltar.

A concrete next step is to connect this predictive substrate with ScienceIDE [11] and ScienceBuddy [34]. ScienceIDE provides executable scientific environments in which agents can run analyses, invoke simulations, and obtain verifiable feedback. These environments could support the collection of action-conditioned experience and the evaluation of decisions informed by learned world models. ScienceBuddy provides an interactive research workspace and couples harness evolution with model learning. Integrating predictive models into this workflow would allow agents to translate researcher questions into candidate experiments, use latent predictions to prioritize them, and compare their outcomes with computational or physical evidence. The resulting

experience could inform subsequent updates to the predictive model, agent policy, and research procedures.

Realizing this loop requires progress beyond predictive accuracy. Key questions include how to calibrate uncertainty under distribution shift, determine when a learned model is reliable enough to guide an experiment, and revise predictive factors as new evidence reveals missing structure. Orthogonal factors do not by themselves establish causal mechanisms, and a shared predictive architecture does not guarantee aligned representations across domains. Future evaluation should therefore measure the quality and efficiency of experiment selection, externally validated hypothesis revision, and transfer to subsequent investigations. The long-term goal is a scientific learning loop in which better world models enable more informative experiments, and those experiments improve both the models and the agents that use them.

1 **CLAVE** **Las limitaciones que reconocen los autores** (las que busca la Guía B): calibrar la incertidumbre bajo cambio de distribución; saber cuándo el modelo es confiable; **revisar los factores** cuando aparece estructura faltante; los factores no establecen mecanismos causales; y una arquitectura común no garantiza representaciones alineadas entre dominios.

2 **PREGUNTA** "Revisar los factores predictivos cuando nueva evidencia revela estructura faltante" es exactamente refinar o ampliar (C4, C7). Los autores lo dejan abierto: es un hueco que podemos tomar.

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A. Approximate Pseudoinverse-Synthesis Stability

The exact orthogonal decomposition in Proposition 1 has a continuous counterpart when the learned analysis matrix is only approximately orthogonal and synthesis uses the pseudoinverse in Equation (5).

Proposition 2 (Approximate stability). *Suppose $P \in \mathbb{R}^{d\times d}$ satisfies $\|P^\top P - I_d\|_2 \le \epsilon < 1$. Then*

$$\sqrt{1-\epsilon} \le \sigma_i(P) \le \sqrt{1+\epsilon}, \qquad \kappa_2(P) \le \sqrt{\frac{1+\epsilon}{1-\epsilon}}. \tag{12}$$

In particular, P is invertible. For exact factor coordinates $u = P^\top z$ and predicted coordinates $\widehat{u} = u + e$, pseudoinverse synthesis obeys

$$\widehat{z} = (P^\top)^\dagger \widehat{u} = z + (P^\top)^{-1} e, \qquad \|\widehat{z} - z\|_2 \le \frac{1}{\sqrt{1-\epsilon}} \|e\|_2. \tag{13}$$

Proof. The eigenvalues of $P^\top P$ lie in $[1-\epsilon, 1+\epsilon]$, which gives Equation (12) and implies $\sigma_{\min}(P) \ge \sqrt{1-\epsilon} > 0$. Hence P is invertible and $(P^\top)^\dagger = (P^\top)^{-1}$. Substituting $\widehat{u} = P^\top z + e$ gives the identity in Equation (13); the bound follows from $\|(P^\top)^{-1}\|_2 = 1/\sigma_{\min}(P)$. □

Thus pseudoinverse synthesis has no reconstruction bias for exact factor coordinates, while the measured orthogonality residual controls amplification of factor-prediction error.

B. Supplementary Reproducibility Information

B.1. Implementation Details

The APEBench Burgers and Kuramoto–Sivashinsky models use an MLP with a first linear-layer weight of shape $[256, 64]$ and OPF dimensions $d = 128$, $K = 4$, and $r = 32$. The Norman single-cell model uses $d = 512$, $K = 4$, and $r = 128$, and defines the prediction target as $z_{\text{response}} = z_{\text{observed}} - z_{\text{control}}$.

B.2. OPF Coefficients

Table 10 reports the dimensions and three additive OPF coefficients in Equation (10) for the listed experiments. Domain-specific terms remain part of $\mathcal{L}^{(\delta)}_{\text{base}}$. The dimensions agree with Table 2.

1 CLAVE **Proposición 2.** Si $\|P^\top P - I\|_2 \le \varepsilon$, la amplificación del error está acotada por $1/\sqrt{1-\varepsilon}$. Con la tabla 5 ($\kappa_2 \approx 1.00005$), en la práctica no hay amplificación.

2 EJERCICIO Deriva (12): los valores propios de $P^\top P$ están en $[1-\varepsilon, 1+\varepsilon]$ y $\sigma_i(P) = \sqrt{\lambda_i(P^\top P)}$. Luego, $\kappa_2 = \sigma_{\max}/\sigma_{\min}$.

3 EXTRA Detalles de implementación: se pueden saltar.

Table 10 | OPF dimensions and coefficients of the three additive regularizers in Equation (10).

Experiment	d	K	r	λ_{orth}	λ_{fac}	λ_{enc}
CITRIS Interventional Pong	160	5	32	.10	.05	.02
CausalWorld state/control	128	4	32	.10	.05	.02
DMC pixel dynamics	128	4	32	.10	.05	.02
DMC-VB held-out episodes	128	4	32	.10	.05	.02
PDEBench Burgers	128	4	32	.10	.05	.02
PDEBench reaction–diffusion	128	4	32	.10	.05	.02
PDEBench shallow water	128	4	32	.10	.05	.02
Synthetic PDE	128	4	32	.10	.05	.02
Synthetic pixel dynamics	128	4	32	.10	.05	.02
WeatherBench2	128	4	32	.10	.05	.02
APEBench Burgers / Kuramoto–Sivashinsky	128	4	32	.10	.05	.02
Hopper-v5, 20-step	32	4	8	.02	.10	.02
Walker2d-v5, 20-step	32	4	8	.02	.10	.02
HalfCheetah-v5, 20-step	32	4	8	.02	.10	.02

1 **PREGUNTA** **Tabla 10:** $K = 4$ en todos los experimentos salvo Pong ($K = 5$), con los mismos coeficientes en casi todos. El número de factores no se adapta al dominio, contra lo que sugiere la introducción (C4).

B.3. AI Use Statement

OpenAI Codex was used for language editing and translation; LaTeX restructuring; consistency, bibliographic, and mathematical checks; cropping and layout of manuscript figures; and feedback on experimental design, statistical reporting, and the interpretation and presentation of results. The authors reviewed all proposed changes and checked cited sources, derivations, and experimental claims against the original materials. The authors made the scientific decisions and take responsibility for the final manuscript.

B.4. Ethics Statement

The clinical experiment uses de-identified UK Biobank participant data accessed through its controlled research and data-governance process. Analyses were conducted within the scope of an approved research application and applicable institutional ethics and data-use requirements. No participant-level data are redistributed.

B.5. Reproducibility Statement

Section 2 specifies the additive OPF loss, its combination with the original implementation loss, the activity statistics, and the synthesis map. Section 3 identifies the data, matched-baseline controls, and metrics, while Table 2 records context–target construction, factor dimensions, evaluation splits, and seed counts. Appendix Table 10 reports the OPF regularization coefficients for the listed experiments. Experiments use the default architecture and optimization settings of the cited backbone implementations, with non-OPF settings shared within each matched comparison. Tables report absolute metrics where scales are comparable, and figures state the aggregation used for cross-task summaries.

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Organizations

[1] PhAI Labs.

[2] The Chinese University of Hong Kong.

[3] Fudan University.

[4] City University of Hong Kong.

[5] University of Bristol.

[6] Stanford University.

[7] University of Oxford.

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